

AN ARCHIVAL ALMA SEARCH FOR UNRESOLVED SPECTRAL-CARRIER TECHNOSIGNATURES
TOWARD STARS WITHIN 40 PC AT 114–873 GHzGLENN J. WHITE^{1,2} AND ROBIN DEY³

ABSTRACT

Radio technosignature searches have concentrated below about 30 GHz, and none above it has reached a nearby star. The ALMA archive reaches it incidentally: observations of discs and planet formation record a spectrum of the star at each field centre, which a uniform reprocessing searches for a drifting carrier — 402 fine-channel spectral windows toward 60 stellar systems within 40 pc, between 114 and 873 GHz. Of 56 threshold crossings, four arise in one execution block that fails a quality criterion fixed in advance; of the rest, 21 coincide with a catalogued molecular transition, including β Pictoris’s carbon monoxide, recovered blind. The unattributed remainder is no more numerous than chance predicts, a comparison that could not see a population below about 11 events. Every unattributed crossing was tested for recurrence in the star’s own frame, over every further window covering its frequency and every drift searched; none returned, including the one ahead of all 512 control positions of its own field. That excludes a carrier holding its frequency inside the window searched and radiating through both observations, not a persistent transmitter of any kind. Injected carriers are recovered nine times in ten at 1.6×10^{15} W for the median system. The 47.7 GHz of union bandwidth searched is about twice the band recurrence could be tested over. The same data’s coarser channels limit an unresolved spectral excess, without discriminating a carrier, down to 90 GHz. Molecular-line confusion and sparse repeat coverage, not thermal sensitivity, limit a search of this kind.

Subject headings: extraterrestrial intelligence, radio lines: stars, submillimetre: stars, surveys, astrobiology

1. INTRODUCTION

The public ALMA archive holds thousands of hours of millimetre and submillimetre observations taken for other reasons, mostly studies of dust discs and planet formation. Their visibilities allow a spectrum at the position of the star at the centre of each field, at a resolution and depth set by the original observer’s correlator setup. This paper searches those spectra for a narrow-band transmitter at the star’s own position. Nothing of the kind survived.

Searches for *technosignatures* — detectable imprints of technology on the electromagnetic spectrum, intended or not (Sheikh 2020; Vidal et al. 2026) — have examined many thousands of stars over six decades, yet sample only 6×10^{-18} of the space of transmitter position, power, frequency, bandwidth and repetition rate (Wright et al. 2018), and essentially all of that effort lies below ~ 30 GHz. No published radio technosignature search above that frequency has reached a star within 40 pc, and few searches at any frequency reprocess existing interferometric archives (White et al. 2026). The band is attractive because a carrier arrives less distorted and a given aperture delivers more gain per watt supplied, and hard because archived correlator channels are far coarser than a dedicated narrowband backend and are crowded with molecular lines — a dense background of real astrophysical signals that a threshold search must separate

from an artificial one (§2).

The signal sought is an *unresolved spectral carrier*, emission confined to a few channels, unresolved at the stellar position, and persistent enough to recur between epochs. Each qualifier is a restriction. A carrier broader than the channel width would not be recovered, nor one that fills the synthesised beam, nor one that appears once and never again — and it is recurrence rather than brightness that separates an interesting event from a detection (Sheikh et al. 2021). The frequency is allowed to drift, as it must for a transmitter carried on a rotating, orbiting body, and the drift searched is bounded by a line-of-sight acceleration rather than by a frequency (§4.1). The channels used are 15.3–1953 kHz wide, fine only relative to this archival sample, so “narrowband” here describes the assumed transmitter and never the resolving power of the data. The targets were chosen by other programmes for other purposes, so every statistical statement that follows is about an archive-selected sample (§6.3).

2. BACKGROUND

Radio searches for technosignatures began at centimetre wavelengths and have largely stayed there. The first targeted experiments (Cocconi & Morrison 1959; Drake 1961), the programmes reviewed by Tarter (2001) and Project Phoenix (Backus & Project Phoenix Team 2002) all worked below a few gigahertz. The modern generation divides into targeted and commensal surveys at centimetre and metre wavelengths (Zhang et al. 2020; Czech et al. 2021; Li et al. 2026; Johnson et al. 2023; Choza et al. 2024), several of them riding back-ends built for other science (Morrison et al. 2023), with wide-field low-frequency coverage from the MWA (Tremblay & Tin-

¹ School of Physical Sciences, The Open University, Walton Hall, Milton Keynes, MK7 6AA, UK; glenn.white@open.ac.uk

² RAL Space, STFC Rutherford Appleton Laboratory, Harwell Campus, Didcot, OX11 0QX, UK

³ VBRL Holdings, 50/5 Huay Kaew Road, Chang Phueak, Chiang Mai, Thailand

gay 2024) and single-dish C- and K-band work at 6 and 18 GHz with the Sardinia Radio Telescope (Manunza et al. 2025). Machine-learning interference rejection is now central to that effort (Ma et al. 2023; Jacobson-Bell et al. 2025; Pardo et al. 2025). The classical threshold search used here lacks that capability, and instead uses a spatial test on the star against control positions in the same field, which a single dish cannot make. A narrowband search also reaches only one class of signal: broadband planetary-scale leakage (Garrett 2026) is outside it by construction.

Above about 30 GHz the record is nearly empty, and that is what makes this frequency range new. ALMA observes in Bands 3 to 10, spanning roughly 84 to 950 GHz; of those eight bands, one has been searched for technosignatures. For continuity with the centimetre literature we note one familiar power scale: the Arecibo planetary radar radiated 2.0×10^{13} W at 2.38 GHz (Sheikh et al. 2025). It carries no transmitter model into the millimetre band, where an aperture of that diameter is not realisable, so the illustrative benchmark of §6.3 is instead a dish of 12 m. Neither is a privileged scale; the limits of this paper are stated as EIRP and need no transmitter model.

Mason et al. (2025) is the only prior millimetre search and the only comparable archival one. They searched Band 3 windows toward 28 Gaia DR3 stars lying in the fields of 4 ALMA calibrators, observed for other science and so usable for free. Their nearest target is at 1.01 kpc, where a 5σ flux threshold corresponds to $\text{EIRP}_{\min} > 6.9 \times 10^{17}$ W. That target lies $25\times$ further away than the most distant star searched here for a carrier and $277\times$ further than the nearest of those: the two samples differ by 1.4–2.4 orders of magnitude in distance, and at a fixed received flux by the square of that, which is 3–5 orders in minimum detectable EIRP. An EIRP ratio between the two surveys would therefore be almost entirely the d^2 term. The quantity that does not depend on distance is the minimum detectable *received* flux, $F_{\min} = S_{\min} \delta\nu$. It must be quoted at a matched channel width, because $S_{\min}$ falls as $\delta\nu^{-1/2}$ while $F_{\min}$ rises as $\delta\nu^{1/2}$. Mason et al. reach 3.8×10^{-23} – 5.8×10^{-23} W m $^{-2}$ across their four fields, on a 30.52 kHz channel at a bare 5σ flux threshold. On that same channel width and the same threshold convention, the fine-channel windows searched here reach 5.2×10^{-24} W m $^{-2}$ at best, a factor 9 deeper, and 7.1×10^{-23} W m $^{-2}$ at the median, a factor 1.5 shallower. The two experiments are therefore comparable in depth and differ in what they cover: distant stars in four calibrator fields, against 60 stellar systems within 40 pc searched for a drifting carrier, over a frequency range reaching about nine times higher. They also identified the obstacles that shape the present method: drift rates scale with frequency, smearing erodes sensitivity unless drift is searched for, and the dense forest of molecular lines raises the confusion background. Every window reported here was re-reduced and re-searched from the raw archive data with the pipeline of §4, so for any star in common the numbers to cite are this paper’s.

Figure 1(a) places this survey among published searches, in minimum detectable EIRP against distance. The vertical axis is physically commensurable: for a car-

rier narrower than every programme’s channel, the minimum detectable EIRP is a total radiated power and not a spectral power density, so the same quantity is plotted for each programme. What differs is the convention, so both are drawn. Every published limit shown is a nominal 5σ threshold at which recovery is not stated, and this survey’s filled symbols are the same thing measured the same way; the open symbols above them are EIRP_{90} , a 90 per cent recovery through the complete automated selection, which is the stricter statement and the one every limit in this paper is quoted on. Any EIRP advantage toward the nearest systems is a d^2 effect and not an instrumental one. Panel (b) shows the other axis that matters. Archived correlator channels here span 3.3 orders of magnitude in width, from 15.3 kHz to 31.25 MHz, and all of them are far coarser than a dedicated narrowband backend, whose 2.8 Hz channels open the full seven orders only when they are set beside ours. The distinctive feature of this survey is therefore its frequency regime and its channelisation, not its depth. It also samples far fewer systems and far fewer epochs than a dedicated search, and complements those programmes rather than deepening them.

A non-detection becomes a quantitative statement only through a stated search volume. Wright et al. (2018) formalised that volume, and Margot et al. (2023) its conversion into an upper bound on the fraction of stars hosting a transmitter; Sheikh et al. (2025) asked where Earth’s own technosignatures would be detectable. Such a bound is valid only if the sample represents the population it is meant to describe. This sample was assembled by other observers for other purposes, so it does not, and §3 states what it is instead.

3. THE SAMPLE

This search covers 1651 spectral windows toward 89 stars in 82 systems within 40 pc, drawn from 404 public ALMA execution blocks and 4.1 TB of raw data through one frozen pipeline, 280 h of telescope time and 1094 h summed over windows. We call that set the *primary census*, and it is the set every limit, every crossing and every expectation in this paper is computed over. Four further sets of blocks enter the paper — a hold-out reserved before any block on its own side of the rule was searched, an archival extension searched afterwards, an out-of-sample control beyond 40 pc, and an external sample used only to calibrate the spatial statistic — and none of them is part of the primary census. The primary census is 404 execution blocks, 1651 searched windows, 89 stars in 82 systems and 1094 window-hours; the hold-out is 77 blocks and 315 windows, and the archival extension 115 blocks and 463 windows. Table 6 gives all five sets in rows that sum, with the results each one enters; wherever a number in this paper is computed over something other than the primary census, that table’s name for it is used at the point of use.

Three words are used consistently and are not interchangeable. A *star* is an individual stellar component; a *system* is a gravitationally bound group, counted once however many of its components were observed; a *target* is one ALMA pointing. The census holds 89 stars in 82 systems.

The census is two experiments, not one. 402 windows toward 65 stars in 60 systems are fine-channel,

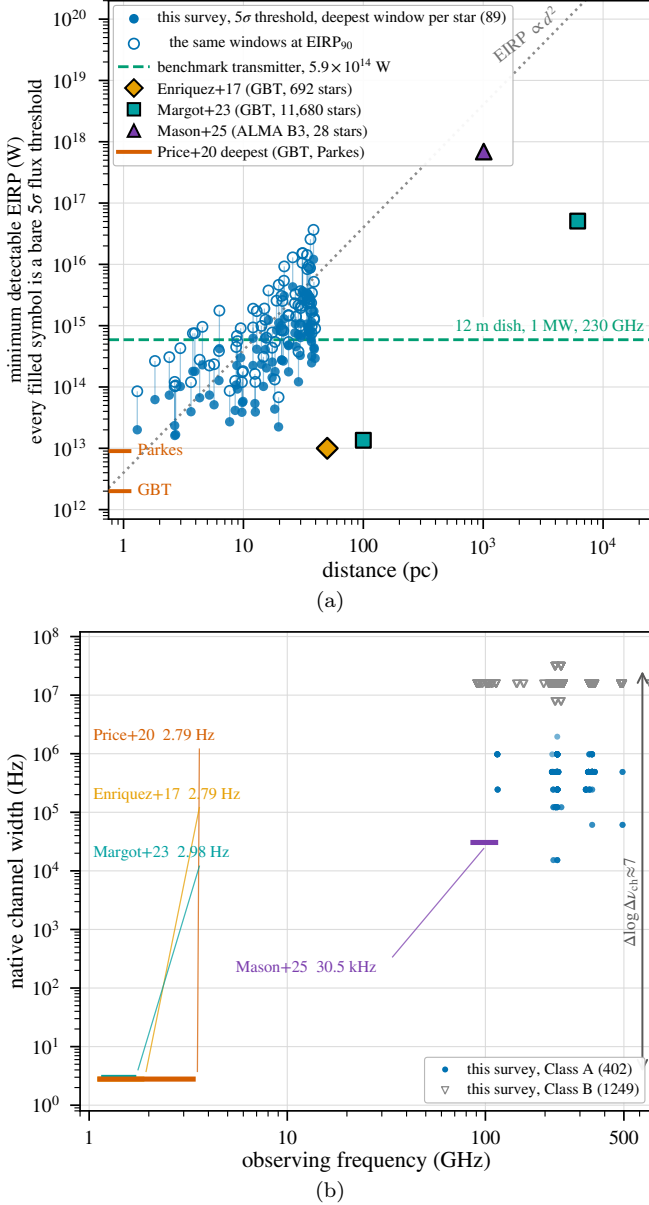


FIG. 1.— Where this survey sits among published radio technosignature searches. **(a)** Minimum detectable EIRP against distance. Blue, this survey, the deepest window per star: *filled*, that window’s bare 5σ threshold, which is the quantity the other programmes quote; *open*, the same window at EIRP₉₀, joined to its own filled symbol. Orange, Enriquez et al. (2017); aqua, Margot et al. (2023); violet, Mason et al. (2025); left-edge ticks, Price et al. (2020), all at their own 5σ thresholds. Grey dotted, $\text{EIRP} \propto d^2$ at constant received flux; green dashed, the illustrative 12 m benchmark transmitter of §6.3. The ordinates are commensurable as total powers for a sub-channel carrier, and the filled symbols make the comparison on one convention; the gap to the open symbol is what completeness costs. **(b)** Native channel width against observing frequency for every searched window (filled, Class A; open, Class B), with the comparison programmes as horizontal segments. This survey’s own widths span 3.3 orders of magnitude; the $\Delta \log \Delta \nu_{\text{ch}} \approx 7$ annotation marks the span that opens once the comparison programmes’ channels are set beside them: the channelisation, not the plotted power, is what differs most between programmes.

called *Class A* throughout: these resolve a drifting carrier and are the primary experiment, and every sensitivity and every exclusion quoted here refers to them unless stated otherwise. The remaining 1249 windows are coarse-channel, *Class B*, and support a secondary search for unresolved spectral excess; they can register a spike but cannot distinguish a drifting signal from a stationary one.

The inclusion criteria are objective and were applied by query. A star enters if its Gaia DR3 parallax is at least 25 mas, measured to better than 20 per cent, and if its proper-motion-corrected position at the observation epoch falls inside the quoted field of view of at least one public execution block. Distances are direct inversions, $d = 1/\varpi$, and the star need not sit at the pointing centre. Disc, planet and spectral type enter as descriptive metadata only, and no star or window was excluded by an unlisted criterion or for short integration. The parallax criterion never binds: the worst fractional parallax error among the stars searched is 2.5 per cent, an order of magnitude inside it, and inside the stricter $\sigma_\varpi/\varpi \leq 0.1$ used in Table 1 to select a reference census on the same footing. The query returns 168 work-list entries; the quoted field of view is optimistic, and only 115 of those stars are in fact contained by a public field, of which 89 yielded a searchable window. The per-star and per-window ledger, including the five-criterion decision tree and every edge case, is in the data release.

3.1. The archival selection

This sample was not designed. It is whatever previous ALMA programmes happened to observe, for their own reasons, and 90 per cent of the stars searched were targeted under a debris-disc or planet-formation proposal. The resulting bias is strong and measurable, and Table 1 tabulates it against a volume-limited reference census selected on the same parallax quality.

The dominant selection effect is distance. Stars within 5 pc are over-represented by a factor 49, and the enhancement falls monotonically with distance to a factor of two *under*-representation beyond 30 pc. Spectral type comes second: A stars are over-represented by about 23 and F and G stars by factors of two to three, because those are the stars with bright debris discs. M dwarfs are mildly depleted — 41 of the 89 searched stars are M dwarfs, a factor 0.8 of their share of the reference census, and only 29 of them carry the fine-channel coverage that can resolve a drifting carrier at all — and that is a weaker statement than the table’s early-type ratios, not a stronger one. Both figures are bounds in the same direction: Gaia temperatures are missing for about half the reference census and missing preferentially for the coolest objects, so the census M fraction is a lower limit and every early-type ratio is therefore a lower limit too. Coverage is also concentrated: 285 of the 1651 windows come from 3 systems. Every statistical statement in this paper is therefore a statement about an archive-selected sample of nearby stars, and none of them is an occurrence rate for the solar neighbourhood. The systems a targeted search would itself have chosen — every sample system within 15 pc with a confirmed planet — are listed in the data release, where 6 of the 15 carry no Class A coverage at all and so no drift-resolving limit: the archival selection acting on exactly the targets a transmitter search

TABLE 1

SELECTION FUNCTION OF THE SEARCHED SAMPLE AGAINST A VOLUME-LIMITED REFERENCE CENSUS, BOTH SELECTED ON THE SAME PARALLAX QUALITY: GAIA DR3 SOURCES WITH $\varpi > 0$, $\sigma_{\varpi}/\varpi \leq 0.1$ AND $1/\varpi \leq 40$ PC (17 566 OBJECTS, 8594 OF THEM CARRYING A `TEFF_GSPPHOT` ESTIMATE), AGAINST THE 89 STARS WITH AT LEAST ONE SEARCHED WINDOW. THE SEARCHED STARS ARE MEASURED FAR BETTER THAN THE CUT REQUIRES — THE WORST FRACTIONAL PARALLAX ERROR AMONG THEM IS 2.5 PER CENT — SO THE TWO COLUMNS ARE SELECTED ALIKE, AND LOOSENING THE CUT TO 0.2 ADDS NO CENSUS OBJECT AT ALL. 86 OF THE 89 SEARCHED STARS ARE CLASSIFIED: 53 FROM THE GAIA TEMPERATURE, 25 FROM THE SIMBAD SPECTRAL TYPE WHERE THAT TEMPERATURE IS ABSENT, AND 8 IDENTIFIED AS WHITE DWARFS; 3 CARRY NO CLASSIFICATION IN EITHER SOURCE.

THE SIMBAD ROUTE MATTERS BECAUSE `TEFF_GSPPHOT` IS MISSING PREFERENTIALLY FOR COOL STARS, WHICH IS THE POPULATION THE M DWARF STATEMENT IS ABOUT. “RATIO” IS THE SAMPLE COLUMN FRACTION OVER THE CENSUS COLUMN FRACTION; > 1 IS OVER-REPRESENTATION. *The strongest selection effect here is distance, not spectral type.*

Property	Census	Searched	Ratio
Distance			
$d \leq 5$ pc	48 (0.3%)	12 (13.5%)	49.3
5–10 pc	240 (1.4%)	13 (14.6%)	10.7
10–20 pc	2069 (11.8%)	18 (20.2%)	1.7
20–30 pc	5391 (30.7%)	19 (21.3%)	0.7
30–40 pc	9818 (55.9%)	27 (30.3%)	0.5
T_{eff} class ^a			
A	24 (0.3%)	5 (6.4%)	23.0
F	402 (4.7%)	9 (11.5%)	2.5
G	860 (10.0%)	16 (20.5%)	2.0
K	1400 (16.3%)	7 (9.0%)	0.6
M	5908 (68.7%)	41 (52.6%)	0.8
Other axes			
Confirmed planet hosts ^b	26 (15.5%)	23 (25.8%)	1.7
White dwarfs ^a	—	8 (9.0%)	—
Searched stars / systems	—	89 / 82	—

^aTemperature bins, not MK types: A ≥ 7500 , F 6000–7500, G 5300–6000, K 3900–5300, M < 3900 K, normalised over the 78 classified non-degenerate stars. The 8 white dwarfs are kept apart because a degenerate is not an early-type star and its photometric temperature would place it among them. Gaia `teff_gspphot` is unavailable for about half the reference census, preferentially for the coolest and faintest objects, so the census M fraction is a lower limit and every earlier-type ratio in this panel is correspondingly a lower limit; the searched column has no such gap.

^bBoth columns are the same positional join, of the 168-entry ALMA-covered work list and of its searched subset, to the NASA Exoplanet Archive `pscomppars` table accessed 2026 October 4; a confirmed planet is a row of that table. The 23 searched hosts carry 50 confirmed planets between them, and 21 of the 23 survive the deuterium-burning mass limit. The Gaia reference census carries no planet information, so the census column in this row is the work list rather than the 17 566-object Gaia census.

would pick.

3.2. Frequency coverage and archival scope

The Class A experiment covers $\Delta\nu_A = 47.7$ GHz of unique sky frequency, and that coverage is sparse: it is 26 disjoint islands scattered between 114.0 and 873.1 GHz, not a band. The endpoints are a range over which observations exist, never a range that was covered. Adding the coarse-channel windows brings the total archival coverage to 118.1 GHz in 46 intervals, of which 113.1 GHz survives the molecular-line mask of §4. The distribution is shown in Fig. 2; a transmitter at a frequency between the islands was never observable here, whatever its power.

The archive is not exhausted either, and the shortfall is accounted for rather than estimated. 177 further blocks lie inside the same query but were not processed when

the census was frozen; Table 6 gives what became of each. 144 of them have since been calibrated and searched, and they enter this paper only as repeat epochs. The largest single gap is the 18 blocks for which the archive holds raw visibilities that were never pipeline-calibrated, 237 GB in all; no effort on our side would close it.

4. DATA AND METHODOLOGY

The search looks for one kind of signal: an unresolved carrier at the star’s own position, narrow compared with a single correlator channel, and drifting in sky frequency because the transmitter accelerates along the line of sight. Over a long track its power is spread across many channels, so a search that treats each channel on its own will miss it. The analysis de-drifts the data before it measures them.

4.1. The de-drifted search statistic

Each window’s dynamic spectrum is stacked along every trial drift rate $\dot{\nu}$. The search statistic is the largest stacked signal-to-noise ratio anywhere on the resulting channel $\times$ drift grid,

$$T(\mathbf{x}) = \max_{\dot{\nu}, \nu} \frac{\sum_t I(\mathbf{x}, t, \nu + \dot{\nu}t) / \sigma^2(t, \nu)}{[\sum_t 1 / \sigma^2(t, \nu)]^{1/2}}, \quad (1)$$

where $I(\mathbf{x}, t, \nu)$ is the spectrum at sky position $\mathbf{x}$ in integration t and channel ν , and $\sigma(t, \nu)$ is the noise scale there. Equation 1 is the matched filter for a constant-drift carrier: it adds the signal coherently along the track and divides by the noise of the stacked spectrum, so T is an absolute signal-to-noise ratio and not a local contrast. The noise scale is the median absolute deviation across 512 control positions at each integration and each channel, scaled to its Gaussian-equivalent standard deviation — local in time and frequency, global in position, shared by the star and every control, and with the across-position median never subtracted from the numerator. The statistic is evaluated at one position at a time, identically at the star and at each control, which is what makes the two comparable. T_* denotes its value at the star.

Spectra are extracted at the star’s apparent position, never from images. For each integration and each native channel the pipeline forms the weighted vector average of the phase-shifted visibilities, in total intensity: the mean $I = (XX + YY)/2$ of the two parallel-hand correlations. That average is the dirty-map value at the extraction position — an amplitude, not a deconvolved flux density, so in a field with extended emission it is not a point-source measurement. The data are used as the archive delivered them, through ALMA’s own calibration and quality assurance, with their flagging carried through, every delivered channel searched, and no regridding, deconvolution or edge trimming. A per-integration median along the *frequency* axis removes the continuum; no median is taken along time, so a one- or two-channel feature cannot be removed by it. One subtlety matters for every power quoted below: the delivered channels are smoothed, so a channel’s effective noise bandwidth is not its separation and flux density times channel width is not yet a radiated power. A response correction computed from the correlator’s own channel shape supplies

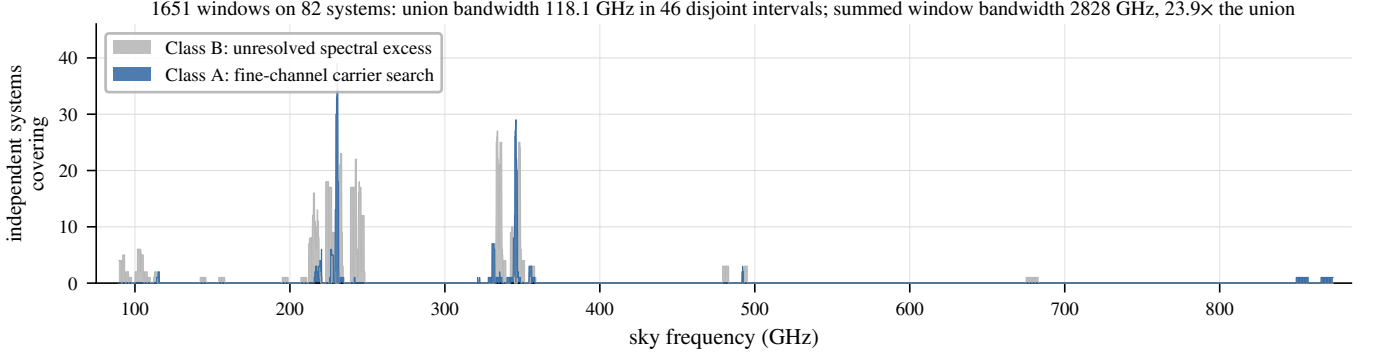


FIG. 2.— Where this survey searched. *Lower panel:* each searched window is a horizontal segment on its host system’s row, rows ordered by distance and labelled alternately on the left and right axes; heavy blue marks the fine-channel carrier search (Class A) and light grey the coarse-channel spectral-excess search (Class B). *Upper panel:* the number of systems covering each gigahertz of sky frequency; the threshold distribution is Fig. 1(a). The coverage is narrow, clustered and repeatedly revisited: 46 islands concentrated near 230 and 345 GHz, with summed window bandwidth 23.9 $\times$ the union. The union measures frequency coverage, the summed bandwidth integrated search effort.

the difference, derived and validated against the effective resolutions ALMA reports in Appendix A.

One convention is fixed here and holds everywhere in the paper. The search runs on the topocentric sky frequencies ALMA delivered, and every drift rate quoted is the apparent one at the telescope, since that is what the data contain and what a reader can recompute from the released channel axes. Two steps need the star’s own frame instead, and both take it: a crossing is compared with the line catalogue, and a repeat epoch is read at the predicted cell, after each block’s frequency axis has been corrected for the Earth’s motion at that epoch and then for the star’s systemic velocity. Nothing here is evaluated in a barycentric frame alone.

Two power quantities carry the main text, both defined in Table 2: the trigger level $P_{\text{trig}} = 4\pi d^2 S \Delta\nu$ at distance d , for a carrier that makes $T_\star$ reach $T_{\text{trig}} = 5$ in one native channel of width $\Delta\nu$, and the sensitivity EIRP₉₀. The second is measured, not assumed: synthetic carriers of known amplitude were injected into calibrated visibilities and recovered through the unmodified pipeline, 8860 of them over 52 block-window configurations of the fine class and 5936 over 35 of the coarse, spanning both ALMA arrays, four bands and the sample’s full range of channel widths. Every limit quoted in this paper is the EIRP₉₀ of its own window, given to two significant figures beside its native channel width; the measured completeness is §5.1 and its error budget is Table 3. The absolute flux scale is checked twice from outside, in two bands on two arrays: the extractor reproduces an independent reduction of the same AU Mic visibilities to 0.994 ± 0.014 , and the published quiescent flux density of ϵ Eridani to 1.10 ± 0.31 (Burton et al. 2022).

The drift range searched is bounded by an acceleration, not by a frequency. Drift rate and line-of-sight acceleration are one quantity in two sets of units, $a_{\text{los}} = c\dot{\nu}/\nu$, so a ceiling expressed as $\dot{\nu}/\nu$ is the same physical limit in every band; the released drift and acceleration ceilings satisfy that relation window by window, to 1×10^{-4} over all 402 Class A windows. The default ceiling is 12.0–13.3 Hz s^{−1} GHz^{−1}, that is $|a_{\text{los}}| \leq 3.60\text{--}4.00 \text{ m s}^{-2}$. That is the acceleration of a transmitter orbiting a solar-type star at a few hundredths of an au. It is the Mason et al. (2025) fiducial with a threefold margin, widened for a confirmed exoplanet host whenever the innermost known planet demands more. It is far narrower than the general

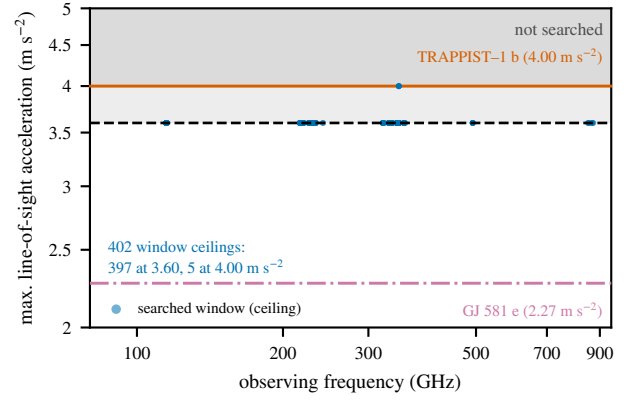


FIG. 3.— What the drift search reaches, as a physical selection function. The drift ceiling of each of the 402 Class A windows (points), converted to line-of-sight acceleration, $|a| = c\dot{\nu}/\nu$, against observing frequency; the shaded region above is unsearched. The boundary is flat because the grid was built at constant acceleration; in the units the search works in it is not, running 1.4 kHz s^{−1} at 115 GHz to 10.5 kHz s^{−1} at 872 GHz. Horizontal lines are reference accelerations: Earth’s rotation at the equator (0.034 m s^{-2}) and orbit (0.0059 m s^{-2}), and the two most demanding known planets of any target, TRAPPIST-1 b (4.00 m s^{-2}) and GJ 581 e (2.27 m s^{-2}). The boundary drawn is the default ceiling, 3.60 m s^{−2}, which is also the median; it runs to 4.00 m s^{−2} on the 5 TRAPPIST-1 windows. For the most demanding case, transiting TRAPPIST-1 b, 29 per cent of the orbital phase lies above the default ceiling and 1 per cent above the widened one.

recommendation of Sheikh et al. (2019), which is set by solar-system rotation and orbital motion rather than by a planet’s own acceleration, and the difference is a deliberate restriction of what this search covers and not a weaker version of their advice. The margin is not cosmetic: at unit margin the search would have been blind to two of the windows that produced a crossing. The grid is uniform in $\dot{\nu}$, and fine enough that a carrier midway between two trials loses no more than 0.26 of a native channel over the longest track. Two limits follow: the rates searched are the apparent ones at the telescope, so a deliberately chirped carrier is not distinguished from an accelerating platform, and a transmitter need not be bound to a known planet at all, so drifts faster than the ceiling are simply unconstrained (Fig. 3).

Only part of the archive can carry a drift search at all, and the two classes reported here are that division (Table 2). Channelisation in the sample is bimodal, with an empty gap between the modes, so the split is unam-

biguous, and it coincides with the physical criterion that matters: whether the maximum searched drift crosses of order one channel or more over the on-source track, $\eta_{\text{drift}} = |\dot{\nu}_{\text{max}}| \tau_{\text{track}} / \Delta \nu_{\text{ch}} \gtrsim 1$. Each window's own η_{drift} , trial-drift count and acceleration ceiling are released catalogue columns, so a reader may reclassify on that criterion instead; the two do not partition the survey identically. **The two classes are physically different experiments**, results are given per class, and they are combined only where that is statistically appropriate.

4.2. From a crossing to a confirmed signal

Every window passes through the same three tests in the same order, and three words name what comes out of them: a *crossing*, an *unattributed crossing* and, had one survived, a *confirmed signal* (Fig. 7). Each is set out below as the quantity it measures, the number that quantity is compared with, when that number was fixed, and what passing it does and does not establish. There is no fourth test, and none of the three is a significance test.

- (i) *Trigger* $\rightarrow$ a *crossing*. The quantity is T_* , the de-drifted statistic of Eq. 1 read at the stellar position; the criterion is $T_* \geq T_{\text{trig}} = 5$ in at least one channel $\times$ drift cell; and both the statistic and the level were fixed before the archival search began. **T_{trig} is a trigger threshold and not a detection significance:** T is not shown to follow any standard distribution under the null, so the level carries no tail probability, and a window holds between 112 and 6.87×10^6 of those cells, so one level buys very different protection in a coarse single-pointing window and in a finely channelised one on a long track. A crossing establishes only that the window goes on to the next two tests.
- (ii) *Line attribution* $\rightarrow$ an *unattributed crossing*. The quantity is the crossing's velocity offset from the nearest transition of a molecular-line catalogue, evaluated in the star's own rest frame; the criterion is $\pm 50 \text{ km s}^{-1}$. The statistic, the trigger and the original species list were fixed before any crossing was classified; the species list was afterwards extended, and that extension is the reason one crossing is reported as unattributed where the list as it now stands would attribute it (Appendix C). The half-width follows the Keplerian line-of-sight velocities of gas surviving beyond a few au, with margin for the systemic velocity. A crossing inside that window is *attributed*, and the attribution is good evidence: a coincidence with a catalogued transition makes an astrophysical origin far more plausible than a transmitter, the more so where a second transition of the same species or known circumstellar emission is present. But a coincidence is a statement about a catalogue rather than a property of a frequency, so this is an attribution and not an exclusion, and **every crossing goes on to the recurrence test whichever side of the window it falls**, wherever a later block covers the frequency. An unattributed crossing is a target for follow-up and not a detection. The window's price is that the survey constrains nothing inside it: $\approx 113.1 \text{ GHz}$ of the 118.1 GHz covered lies outside (Appendix C).

- (iii) *Independent recurrence* $\rightarrow$ a *confirmed signal*. The quantity is T_* at the cell the first epoch predicts, read in another execution block of the same star at a single drift rate after the frame correction of §4.1; the criterion is that it reach the trigger again; and the test was specified before the repeat epochs were searched. It is a weak criterion, and the archive's uneven temporal sampling is why: only part of the sample has epochs separated well enough to apply it at all.

Two diagnostics hang off the second test, and the word is exact: each says which crossing deserves attention first, and neither decides anything. The first ranks T_* against the same statistic at 512 control positions in the same window, drawn once from a fixed seed into an annulus placed by each block's own synthesised and primary beams so that no control sits on the star; its lowest value, $1/(512 + 1)$, means the star beat every control. It is a ranking filter and not a significance test, because the stellar and control positions are not demonstrably exchangeable: out of sample, on 315 windows the frozen pipeline had never seen, the stellar rank has median 0.405 where one half is expected if the two were interchangeable, and a parameter-free correction for the annulus geometry does not restore an exchangeable statistic (Appendix B). It also responds to compactness rather than to origin, as β Pictoris shows: genuine carbon monoxide ranks below its own control ring there, because the resolved disc lifts the ring along with the star. The second diagnostic is a visibility-domain fit, which asks whether a crossing's emission is at the stellar coordinates rather than elsewhere in the field; it cannot discriminate at the trigger either, because a noise maximum at the star has flux at the star (§5.5). **Neither gates a disposition, so neither is charged to the sensitivity**, and the limits here are correspondingly deeper than a chain charging for the rank would give. Throughout, p denotes a genuine significance level, and no probability computed against the rank's own null is called one.

One execution block is not a null, and a rule fixed in advance says so. The comparison with the control positions needs each block's ensemble to behave like noise, and a block fails when the median control statistic over its own windows, scaled by the median of that quantity over its class, exceeds twice that scale. Of the 404 primary-census blocks, 1 fails — A002_Xc7fa6f_X28a8 toward BD+05 1668, at 4.1 where the next worst is 1.12. **The Class A results are identical with it and without it**, the same 402 windows and 52 crossings either way, because its 4 windows are all coarse-channel; what it does hold is 4 of the 56 crossings, the survey's entire Class B crossing population (Appendix D).

Because a fixed threshold is not a fixed false-alarm rate, the number of crossings expected by chance is computed window by window, from the exceedance rate measured at each window's own control positions; the reserved blocks then test the one assumption that makes, which is that a star behaves like a control position in its own field (§5.3.1).

Three corrections precede the statistic: the extraction field is chosen so that the star lies inside the delivered primary-beam field and not merely inside the quoted field of view; the stellar position is propagated to the observa-

TABLE 2

THE QUANTITIES AND THE THREE WORDS THIS PAPER USES, DEFINED ONCE. EVERY TERM BELOW IS USED IN THIS SENSE THROUGHOUT AND IN NO OTHER.

Term	Definition
Crossing	A channel $\times$ drift cell at the stellar position reaching $T_\star \geq 5$. An event requiring investigation; no claimed significance.
Unattributed crossing	A crossing lying more than 50 km s^{-1} in the star’s frame from every transition of the frozen line catalogue. A target for follow-up; not a detection.
Confirmed signal	An unattributed crossing recovered at the predicted stellar-frame frequency and drift in an independent block. The only confirmation criterion.
P_{trig}	The power of a carrier that makes $T_\star$ reach the trigger in one native channel. An internal pipeline level: not a sensitivity, not a significance.
EIRP ₉₀	The power at which nine injected carriers in ten are recovered at the star through the trigger. A measured completeness, not a detection probability.
Class A	The primary experiment: 402 fine-channel windows, median 488.3 kHz, toward 60 systems over 47.7 GHz of union bandwidth, in which the searched drift crosses of order a channel or more.
Class B	The secondary experiment: 1249 coarse-channel windows, mode 15.625 MHz, constraining a persistent unresolved excess. No drift discrimination, and never a carrier search.
The sample	Neither volume-limited nor representative of the solar neighbourhood: the stars ALMA happened to observe for other reasons (§3).

tion epoch for proper motion, though not for the annual parallax, whose omission is corrected window by window in §5.1; and on-source time is accounted per integration from the delivered exposure records rather than from the scan intentions. That the chain finds what it is meant to find is shown twice over, by the injected carriers above and by β Pictoris CO — a real feature at a known stellar position, recovered and localised in two transitions and 9 execution blocks (§5.4).

5. RESULTS

The primary experiment is the Class A search for a drifting, unresolved spectral carrier: 402 fine-channel windows toward 60 systems. The Class B search for an unresolved spectral excess is secondary and carries no drift discrimination: 1249 coarse windows. Together they are the 1651 windows of the released catalogue. **No event satisfied the confirmation criterion set out in §4.2.** What follows is first how deep the search reaches, and then what it found and how each finding was disposed of.

5.1. Sensitivity: the power at which a carrier would have been found

The sensitivity of this survey is a measured completeness and not a threshold. Artificial carriers were added to the real calibrated visibilities at a random sub-channel phase and a random drift from each window’s own grid

and recovered through the unchanged pipeline; the power at which nine in ten come back is EIRP₉₀, the only sensitivity quoted here. *It is measured through the chain the survey applies* — the trigger and localisation at the stellar position. The 512-position spatial rank annotates a crossing rather than deciding it (§4.2) and is not charged against it: it would cost a further factor 1.31 and in 3 bright-ring windows could not be met at any injected power (Fig. 4), and a limit should not be charged for a gate that disposes of nothing.

Measured that way the completeness is one number. Nine carriers in ten are recovered at $3.03 P_{\text{trig}}$, where P_{trig} is the window’s own trigger power, and that class-average factor is determined to ± 2 per cent. Every one of the 49 directly injected Class A windows resolves a ninety per cent point of its own, between 2.41 and $4.80 P_{\text{trig}}$, and the value does not depend on the brightness of the window’s control ring: the 5 windows whose ring carries resolved emission come back at $3.24 P_{\text{trig}}$ against 2.94 for the rest (Fig. 4). One factor therefore serves the class. Class B carries its own factor $4.17 P_{\text{trig}}$, measured the same way from 5936 carriers injected into 35 coarse windows of 14 execution blocks and weighted by the coarse catalogue’s own distribution of ring brightness; it is what sets the limits for Barnard’s Star, Wolf 359 and Proxima.

The resulting limits. The median Class A window reaches $2.9 \times 10^{15} \text{ W}$ of equivalent isotropic radiated power and the median system, on its best window, $1.6 \times 10^{15} \text{ W}$, with a spread across systems of 6.8×10^{13} – $3.7 \times 10^{16} \text{ W}$ (Fig. 5a). The round 10^{15} W level is reached for 25 of them. Table 3 collects the terms acting on those numbers. The calibration of the factor is good to $\times 0.91$ – $\times 1.09$, and that is the interval which applies to the median and to the counts. One bias is not inside it and runs one way only: an injected carrier is added to visibilities that have already been calibrated and so suffers no atmospheric or instrumental decorrelation, whereas a real carrier does, so **every limit in this paper is optimistic by up to 20 per cent** on that account — a figure declared from the literature rather than measured here.

Carriers were injected directly into 49 of the 402 Class A windows, 12 per cent, and the result transferred to the remaining 353, which is the dominant caveat on all of the above. The injected windows are the Class A windows of execution blocks drawn in a stratified random order fixed in advance, without reference to which windows hold a crossing; they hold 7 crossings against the 6.3 the population rate predicts. *The unit is the block and not the window*, because a block’s calibrated visibilities are what an injection campaign has to rebuild: band and array were controlled by the draw, while channel width, ring brightness and integration length vary inside a block and were only compared with the population afterwards, agreeing to 22 per cent on the worst axis.

Every limit therefore rests on the class-average calibration, and what one window may differ from it by is a separate statement. The median ninety per cent point differs by 0.18 between bands, 0.24 between channel widths and 0.20 between the two arrays. The largest of the three is the channel width, and none of them is as large as the gate the chain no longer charges for. The array difference survives setting the bright-ring windows aside — 0.20 without them — so it belongs to the array and not

TABLE 3

WHAT SETS EIRP_{90} , AND IN WHICH DIRECTION. THE UNCERTAINTY ON THE CALIBRATION IS AN UNCERTAINTY ON THE CLASS-AVERAGE FACTOR, AND SO ON ANY STATISTIC FORMED OVER MANY WINDOWS; THE SPREAD BETWEEN WINDOWS IS AN OBSERVED DISPERSION AND IS THE STATEMENT ABOUT ONE WINDOW'S OWN LIMIT; THE SIGNED TERMS ARE BIASES, STATED WITH THEIR DIRECTION RATHER THAN FOLDED INTO AN ERROR BAR. THE CHANNEL RESPONSE IS CARRIED BY THE INJECTIONS RATHER THAN APPLIED AFTERWARDS, SO IT MUST NOT BE APPLIED AGAIN (APPENDIX A).

	value	what it does to the limit
Class-average completeness	$\times 3.03$	nine carriers in ten recovered at this multiple of a window's own trigger power, over 49 injected Class A windows
<i>Uncertainty on that calibration</i> (per cent), combined in quadrature		
The factor itself	± 2	sampling error of the pooled value over the 49 injected windows
Flux, distance and visibility scale	± 9	calibration
Combined	$-9 / + 9$	$\times 0.91 - \times 1.09$ on the factor, and so on the median limit and on the system counts
<i>Spread between windows.</i> An observed dispersion, not an error bar, the 49 injected windows resolve $2.41 - 4.80 P_{\text{trig}}$ individually, so an uninjected window's own completeness is not known to the interval above		
Window to window	$\times 0.80 - \times 1.58$	
<i>Signed terms.</i> A bias is not an error bar and is not added to one.		
Channel response	$\times 1.93$	carried by the injections, so already inside the completeness above the injected profile falls short of the exact response: <i>conservative</i>
its residual	$+8$	
Primary-beam model	$+0 / + 10$	ALMA's beamwidth in place of the pipeline's Airy-null one: nothing in the median window, 4 windows of one star
Decorrelation	$+0 / + 20$	an injected carrier suffers none and a real one does, so every limit is optimistic by up to this much; declared from the literature, not measured here

to its fields, as a compact array keeping more extended emission inside its beam would. Across the injected windows as a whole the ninety per cent point ranges from $\times 0.80$ to $\times 1.58$ of the adopted factor, so an individual uninjected window's own completeness is not known to the precision of the interval that applies to the class.

One geometric term acts window by window, and it is measured rather than modelled. The phase centre assumed in the search omits the annual parallax, which displaces the star by a median 0.022 of a synthesised beam and at most 0.67. Each limit is therefore divided by the amplitude a compact source retains at its own displacement — read off an injection into the same visibilities at both positions, not off a beam model (Appendix A) — which makes the median Class A limit shallower by 1.5 per cent. One window, Wolf 28 in A002_X9f2ff8_X25b0, has no astrometric keys and is left uncorrected.

A carrier that is not always radiating is penalised in proportion. The de-drifted stack is coherent across a whole execution block, so a carrier present for a fraction of one contributes to that fraction of the integrations and comes back at that fraction of its continuous amplitude. Over 3888 trials in 12 windows of 10 stars, at 6 duty cycles within a track down to 10 per cent and at amplitudes between $\times 1.8$ and $\times 145$ of each window's own trigger, the recovered amplitude followed the duty cycle to 1 per cent. A transmitter radiating a tenth of the time therefore needs $10\times$ the power quoted above, and because those trials carry no drift the joint drifting and intermittent case is not measured here at all.

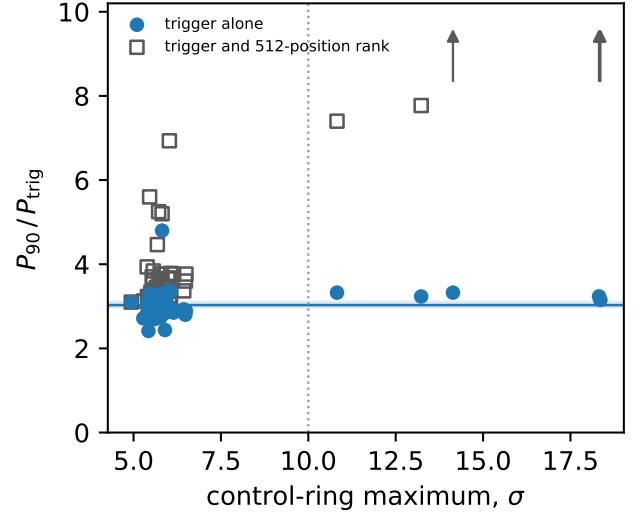


FIG. 4.— Why one completeness factor serves the whole class. Each of the 49 directly injected Class A windows is plotted at the power that recovers nine carriers in ten, against the maximum of its own 512-position control ring. Filled circles are the criterion adopted here — the trigger and localisation at the star — with the pooled value and its uncertainty as the line and band; they show no trend with the ring. Open squares add the demand that the carrier outrank every control position, and arrows mark the windows that then reach ninety per cent at no injected power. Dotted, the 10σ ring brightness above which a ring carries resolved emission rather than noise.

5.2. The nearest stars: Barnard's Star and Wolf 359

Barnard's Star is the second-nearest stellar system and Wolf 359 the third if brown dwarfs are excluded, and to our knowledge these are the first millimetre or submillimetre technosignature limits for either (Enriquez et al. 2017; Price et al. 2020; Margot et al. 2023). Both are non-detections. *Both limits come from the secondary Class B experiment and not from the carrier search:* the archive holds only coarse Band 6 windows toward these stars, so what is constrained is an unresolved spectral excess and not a drifting carrier. On the EIRP_{90} convention of §5.1 Barnard's Star reaches $2.7 \times 10^{14} - 4.0 \times 10^{14}$ W over 12 windows and Wolf 359 $3.1 \times 10^{14} - 4.5 \times 10^{14}$ W over 12.

5.3. Threshold crossings and how each was disposed of

The result is a chain of three tests, and it is read in one direction. Of the 1651 windows, 56 contain a channel $\times$ drift cell at the stellar position reaching $T \geq 5$, and such a cell is a *threshold crossing*. 4 lie in the one execution block that fails the quality criterion of §5.3.2, fixed before it was applied, so the population characterising the search is the 52 crossings in quality-passing data and those 4 are reported beside it rather than within it. The molecular-line mask, $\pm 50 \text{ km s}^{-1}$ wide and evaluated in each star's own rest frame, identifies 21 of the 52 with a catalogued transition and leaves 31 unattributed; every crossing is in one column or the other. Of the unattributed, 1 outrank all 512 control positions of their own window, and neither recurs in any later epoch covering its frequency. **No crossing survives the chain, and this survey reports no detection.** Figure 7 carries the chain with the number surviving each step and Tables 8 and 9 the crossing-by-crossing ledger behind it.

The mask is informative rather than permissive, and its half-width follows from the Keplerian velocity bound

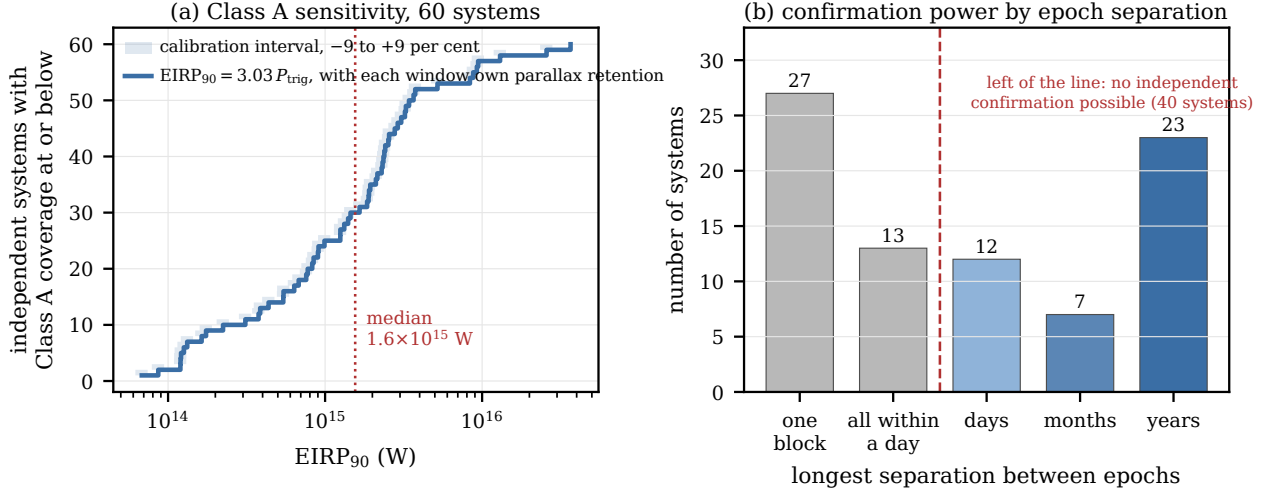


FIG. 5.— (a) Class A sensitivity: the cumulative number of systems whose best window reaches a given EIRP_{90} , the power at which nine carriers in ten are recovered through the trigger at the star. Every window is drawn on the one adopted convention, $3.03 P_{\text{trig}}$ with that window’s own annual-parallax retention applied, and the shaded band is the calibration interval of Table 3. (b) The temporal selection function: systems grouped by the longest separation between their searched epochs, an independent epoch being a separation of more than 1 day. **For the 27 systems observed in a single block, and the 13 whose blocks all fall within one day — left of the dashed line — independent confirmation was not possible at all, whatever the sensitivity in panel (a).**

on circumstellar gas in these systems rather than from tuning (Appendix C). The attributions are kinematic and not an accident of bandwidth, and the comparison that shows it is per window and not over the union band: of the 51 crossings for which the release records a measured peak frequency, 11 fall within $\pm 50 \text{ km s}^{-1}$ of a searched transition against 4.7 expected were each peak free to land anywhere on its own window’s channel grid (Appendix E).

5.3.1. Most of the crossings are expected by chance

The trigger is a threshold on a statistic and not a false-alarm rate: the number of searched channel $\times$ drift cells per window spans 4.8 decades, so one level buys very unequal protection. How many crossings chance should produce is therefore measured window by window, with the same selection on both sides of the comparison — the 402 Class A windows of the census, and in each of them a cell at the stellar position reaching the trigger which the line mask does not attribute. A window’s own 512 control positions measure how often a position in that field reaches the trigger; summed over the windows and reduced by the share of a window’s own channel grid that the attribution windows take, 9.2 per cent, chance predicts 32.8 unattributed Class A crossings and 29 are found. That share is measured on the crossings themselves rather than averaged over the searched band, because the windows are tuned to transitions and a crossing therefore lands inside an attribution window more often than the band average would predict; it lowers the expectation. A further two unattributed crossings, both toward HD 14055, fall in blocks the census windows do not cover and so carry no expectation to be compared with; counting those, the ledger holds 31.

The agreement is real without being sharp. The prediction carries a Poisson scatter of 6 crossings, so a population of genuine signals would have to exceed about 11 events before this count could register it; what excludes each crossing individually is the recurrence test of §5.6.

5.3.2. One execution block accounts for four of the crossings

4 crossings belong to BD+05 1668 (GJ 273, Luyten’s Star), all of them in one ACA execution block, A002_Xc7fa6f_X28a8, whose whole field is above threshold: 512 of 512 control positions exceed $T = 5$ against a survey median control maximum of 2.75. The single-integration noise is as predicted (0.1609 Jy against 0.1611 Jy) but the residual does not integrate down — a jackknife on the channel mean gives 0.030 Jy against 0.0048 Jy thermal — so a slow drift common to every channel and position inflates the statistic across the field, and the 24 sibling executions of the same field, array and spectral setup reach only $T_{\star} = 4.26$ and produce no crossing. *What is inflated is the field and not the star.* The strongest of the four reaches $T_{\star} = 73.4$ where its own control ensemble peaks at 74.7, and between 1 and 421 of the 512 control positions stand above the star in these windows, so there is nothing here to localise and no evidence either way about a carrier. Applied block by block the same criterion fails this block and no other (Appendix D.2), which is why its 4 crossings are reported apart from the 52 in quality-passing data, and Appendix E takes what the block might instead be to the windows in which the star lies well off the phase centre.

5.3.3. Stellar flares are excluded by a broadband prediction

Stellar activity is not excluded by the sample, which contains known flare stars. But a millimetre flare raises every channel of a window alike, so one able to produce a single-channel crossing at the trigger must produce a $5\sqrt{N}\sigma$ continuum event in the same integrations. Over the 48 crossings with a recoverable continuum measurement that prediction runs 49–4,717 σ , against an observed continuum statistic of median -0.29 with $|Z| < 3$ in 45 of them. One block settles it: AU Mic A002_Xa42f75_X319 holds both a crossing ($T_{\star} = 5.27$) and the published millimetre flare of MacGregor et al. (2020), and *that real, bright, independently published flare contributes 0.17 σ to the single-channel statistic* — raising the continuum at $Z = 10.3$ while the crossing’s own de-drifted track

reaches $Z_{\max} = 2.1$.

5.4. β Pictoris: an astrophysical positive control, and the obstacle it identifies

A search that finds nothing must show that it could have found something, and the survey's strongest features show exactly that. The frozen pipeline recovers the β Pictoris circumstellar carbon monoxide blind, without being told the disc is there. It recovers it in two rotational transitions and in 10 independent execution blocks spanning 8.4 yr, and in 9 of those the star also outranks every control: 5 Band 3 windows at T_* up to 30.76, and 4 Band 6 windows at 11.68. The identification is kinematic and does not rest on those statistics. Carried through the frame chain — to the barycentre at each block's own epoch, then to the star's rest frame — all 10 crossings lie within 0.5 km s^{-1} of the stellar systemic velocity, in the two lowest CO transitions of a star with mapped circumstellar CO (Dent et al. 2014; Matr   et al. 2017). The Band 3 peaks agree across blocks to 1.00 channel once each block's own tuning is removed, so any apparent drift is far below the grid searched. The alternative reading requires a transmitter placed by coincidence at matched offsets from two CO transitions of one star.

It is also the control that shows the recurrence test of §5.6 working in the positive direction: every flagged β Pictoris carbon-monoxide feature is recovered in an independent execution block, the screened Band 6 window returning $T_* = 8.95$ in the second block of its own recurrence set. A non-recurrence elsewhere means nothing unless the same machinery recovers a repeat where one exists.

One of those blocks is a counterexample, and it is the more useful half of the control. In it several of the 512 control positions exceed the stellar statistic, so the spatial rank disfavours a window holding carbon monoxide known to be real and centred on the star; the disc is resolved on the scale of the primary beam in that configuration, which is the geometry the statistic is built to reject. **The rank is therefore a test of compactness and not of authenticity**, and on a real extended source it errs conservatively — a stronger statement than a clean positive control, because it bounds the diagnostic's reach as well as demonstrating it (Fig. 10).

HD 48370 is the complementary case and is equally instructive. Its crossing reaches $T_* = 26.83$, but the control ring rises with the star, to 18.36, as emission filling the primary beam does. The velocity matches the prominent CO(2→1) that Cataldi et al. (2023) report toward the source and flag as possible foreground cloud emission. The block's $^{13}\text{CO}(2\rightarrow1)$ window is bright at the ring and faint at the star. Extended line emission is therefore rejected by the same spatial statistic that promotes β Pictoris.

Taken together, these two results identify the dominant obstacle for millimetre technosignature searches, and it is not sensitivity. The strongest apparent signals in this survey are circumstellar and foreground molecular emission, and 21 of the 56 crossings land on a catalogued transition. The mask that removes them costs 4.3 per cent of the unique-frequency union over 12 species, and it is nevertheless the single most productive test in the chain. The warning for future work is about line lists rather than about depth, and Appendix C puts a num-

ber on it.

5.5. The two crossings that lead their own control fields

Two crossings reach a stellar statistic above every one of the 512 control positions in their own window: 61 Vir at 346.313811 GHz and CP-72 2713 at 344.269746 GHz, both in Band 7. They are the most interesting individual events in the search, they are drawn in Fig. 6 and tabulated in Table 4, and neither is a detection. The control rank is recorded for them because it says which crossing was examined first; it disposes of nothing (§4).

CP-72 2713's crossing lies -12 km s^{-1} from SO(8₈-7₇) in the star's own frame and is attributed by the species list, which is applied to it on the same terms as to every other crossing and which attributes further crossings elsewhere to the same molecule (Appendix C); that leaves 1 unattributed crossing leading its own control field. The label disposes of nothing either, and both events go on to the recurrence test.

Neither is present when the same frequency is observed again. 3 other execution blocks cover 61 Vir's cell in the star's own frame; a carrier persisting at the discovery flux would have returned $T = 9.7$ in their combination against 2.71 found, which excludes persistence at 9.3σ . CP-72 2713 is covered by 5 later blocks, of which the deepest began 2.05 hours later and is 1.1 times deeper than the discovery: a persisting carrier would have given $T = 6.2$ there against 3.60 measured, and $T_{\text{pers}} = 6.30$ against the five combined.

The visibility-domain re-measurement sits beside the chain rather than inside it. 51 of the 56 crossings were re-fitted by phase rotating to the star and following the recovered drift, and 41 are consistent with a point source there; but a noise maximum at the stellar position has flux at the stellar position, so at the trigger the fit returns a real part close to the trigger itself. Injected sources in the bin these crossings occupy return $\text{Re}/\sigma = 6.58 \pm 1.97$: both events are consistent with a real source and neither excludes one. The three control sets the experiment uses are different sizes and should not be confused: the rank screen compares the star with 512 image-plane positions, 8 of those retain a full dynamic spectrum, and 12 are reachable in the visibilities. **The discriminating evidence is the line mask and recurrence.**

5.6. Recurrence, applied to every unattributed crossing

A persistent transmitter should be there when the same frequency is observed again. That is the confirmation criterion, and it is applied in the star's own frame: the expected channel follows from the Earth's motion at the repeat epoch and the star's systemic velocity, the discovery drift is carried to it, and the survey's own statistic is read there. Nothing is maximised, so what comes back is an amplitude with an uncertainty. The frame is not a detail: for η Crv it moves the expected channel by -80.2 channels, so a sky-frequency test would have read a channel holding no information about the event.

One channel assumes more than the data support, because a drifting carrier is a carrier that is accelerating. At the largest acceleration measured here, 3.6 ms^{-2} , a body held by a star of $2 M_{\odot}$ or less circles it every 4 days at 176 km s^{-1} , so its carrier need not return to the same stellar-frame channel at all. The window actually

searched at each repeat epoch is therefore not one channel but the whole spectral window at every drift in the grid, $\pm 833 \text{ km s}^{-1}$ about the predicted channel, which spans that excursion for 28 of the 34 crossings whose drift is known. Of 319 covering windows, 291 hold nothing above the trigger anywhere in them at any drift; the 28 that do are no more than the 27 chance predicts in trials that size, each is already a row of Table 8, and the nearest lies 56 km s^{-1} from a predicted channel.

Nothing recurs. All 35 of the 35 unattributed crossings were tested, over 425 covering repeat windows in 132 execution blocks, 32 of them searched after the statistic and the mask were fixed. At the predicted channel the largest statistic over all of them is $+2.00$ at the carried drift and 4.71 over every drift trial, against a trigger of 5 . 34 crossings retain a discovery amplitude and so carry an exclusion, 4 of which are withheld because they would inherit the amplitude of the block that fails the quality criterion of §5.3.2. For each of the 30 that remain the repeat coverage rules out a carrier at the discovery amplitude, for 18 of them one at half that amplitude or less, and in the deepest case at 0.017 of it; read instead as two measurements of one amplitude, discovery and repeat disagree by 3.4 to 5.6σ , median 4.9σ , which is as much as the pair can say when the discovery amplitude is itself a five-sigma measurement. The last crossing, HD 14055 at 331.1559 GHz , retains no discovery spectrum and so has no exclusion, but the criterion needs only a frequency and a drift grid: 15 of the 17 windows covering it hold nothing above the trigger at all, and the 2 that do hold crossings already in the ledger, 500 km s^{-1} away. Table 8 gives the test crossing by crossing, and Appendix B the measured null of the statistic read there.

5.6.1. What can be searched, and what can be confirmed

A sensitivity limit and a confirmation limit are different statements, and this experiment has two extents. It searched 60 systems and 402 Class A windows over 47.7 GHz of union bandwidth; it could have confirmed a signal in 30 of those systems, the ones observed again at the same frequency more than a day apart, over 23.3 GHz or 49 per cent of that union, and in 11 systems over 8.5 GHz beyond a year (Fig. 8). The difference is the cost of retuning.

An independent execution block is not an independent epoch. Every test above uses a block other than the discovery block, but the blocks are not all separate in time: the usable baselines run from 1.22 hours to 1.8 yr, and for 19 of the 34 crossings the nearest covering block falls within a day, while 1 have no covering block beyond one. **What these tests exclude is a transmitter whose carrier stays inside the searched window and radiates through both observations: over the hours to days where there is repeat coverage they constrain persistence, over months and years they bound no duty cycle, and for the 44 systems with no second block covering a frequency the first one covered they bound nothing at all.** A crossing toward one of those could not be retested from the archive at any strength, and the null reported here is a statement about transmitters that would have been seen again.

5.6.2. The reserved hold-out and the control beyond 40 pc

Two sets of execution blocks were searched with the frozen pipeline and then reserved for calibration (Table 6); both hold data on real targets and both are reported here as searches. The hold-out is 77 blocks and 315 windows toward 36 stars, 214 h on source, withheld by a rule fixed before the search finished and sharing no execution block with the census. It contains 8 threshold crossings, the strongest at $T_\star = 8.64$, and *not one outranks its own control positions* — the fewest controls above the star in any of them is 5 of 512. 7 were tested for recurrence on the same code path as the census, and exactly one is present again: HD 285968 at $+9 \text{ km s}^{-1}$ from CO(2–1), returning $T = 5.87$ at the predicted cell of 4 later blocks when the drift is maximised, and $T = +5.49$ at the matched drift, which is the value Table 5 prints. It is a molecular line, and *that it recurs is the test working in the positive direction*; for every other tested crossing the repeat coverage rules out a carrier at 0.47 of the discovery amplitude or less (Table 5).

The control beyond 40 pc is 47 blocks and 198 windows toward 22 background sources, 88 h on source, yielding 8 crossings whose strongest is $T_\star = 12.7$ and none of which outranks its own controls. A systemic velocity is known for only 0 of them, so the set supports no limit and shows only that the pipeline behaves the same way on fields it was not tuned for. The third reserved set calibrates the spatial rank and is not external at all (Appendix B), so the out-of-sample statement this paper makes is the hold-out's.

5.7. A stellar-frame multi-epoch stacked search

Combining the epochs of a star in that star's own rest frame deepens the limit. 58 of the surveyed stars, lying in 58 separate systems, hold more than one searched execution block. Over 315 stacked groups, 2177 epoch-windows and 1426 h, the flux limit improves by a median factor 1.52 over each star's best single epoch and 4.45 at most. 11 further stars stack too — 10 control stars beyond 40 pc and 1 covered only by archival blocks — and are left out of those figures; over all 69 stacked stars the median gain is 1.46, their stacks being shallow two-epoch ones. Every stacked limit is measured by injection into the stack itself and never read off a stacked spectrum as a threshold.

A fixed stellar-frame frequency is not a fixed observed frequency, so the epochs are registered before they are added. The barycentric term alone swings by a median 8.2 km s^{-1} between the epochs of one star and by as much as 53.9 — many channels at the 488 kHz Class A width — so epochs added as observed would dilute rather than reinforce. Each epoch is transformed to the stellar rest frame, divided by its own primary-beam response and combined by inverse variance over the channels of common stellar-frame coverage, keyed on the star's catalogued identity rather than the directory the search ran in. Spectra whose archived visibility weights are pathological are excluded first (27 of 2838, which would otherwise dominate the sum). The transform is verified on a real line: HD 285968's molecular line collapses from a 19.26-channel topocentric spread to 0.798 channels over 4 epochs spanning 44 d.

The criterion was fixed before any stellar position was examined: a stacked $Z \geq 6.0$ above every stacked control maximum, no epoch supplying more than 0.60 of the

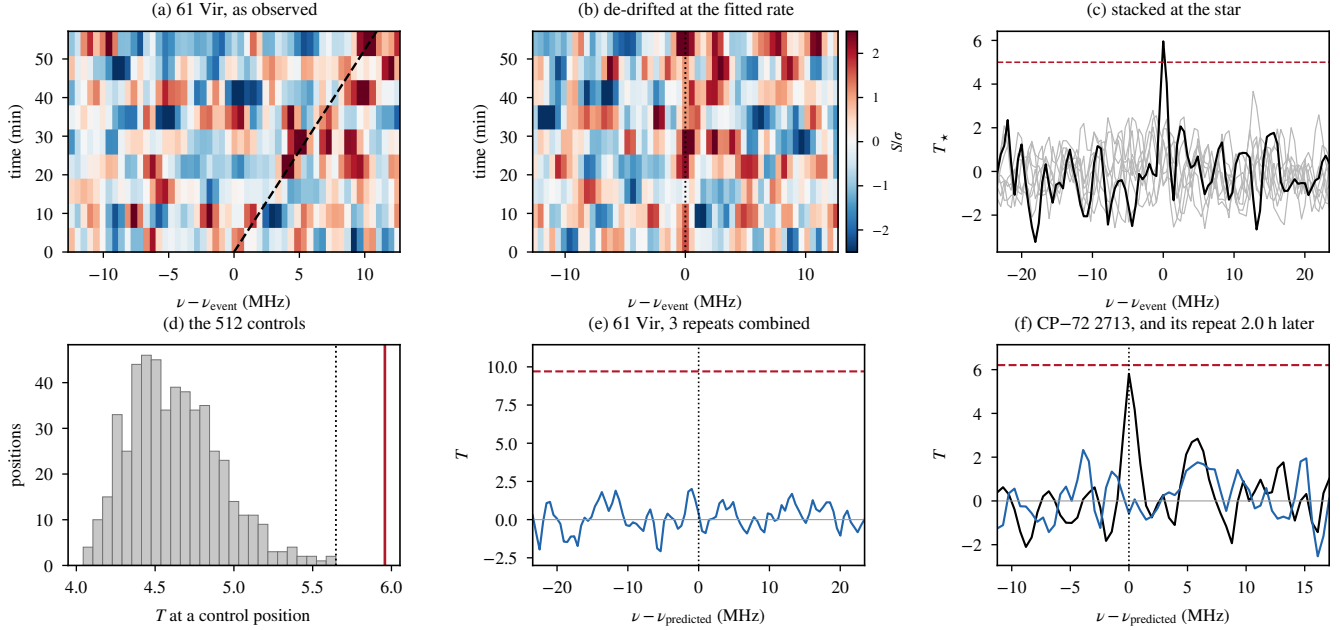


FIG. 6.— **Why this survey reports no confirmed signal although it produced an interesting event.** (a) The dynamic spectrum at 61 Vir’s position around the crossing at 346.313811 GHz, observed 2017 May 20, with the fitted drift of $+3182 \text{ Hz s}^{-1}$ as the dashed track; the 225 integrations are binned into 9 and plotted in units of each bin’s noise. At $119 \pm 20 \text{ mJy}$ against 303 mJy per integration the carrier is 0.39 standard deviations in any one of them and is invisible by eye, which is the point: the event exists only in the 38-minute coherent stack. (b) The same rows shifted by that drift, which carries the carrier 22 channels of 488.3 kHz across the observation; 8 of the 9 bins are positive at the marked channel. (c) The de-drifted stack at the star (black) and at the 8 control positions with retained dynamic spectra (grey), in units of its own noise; dashed, the trigger. (d) The same statistic at each of the 512 rank-screen control positions: the star reaches 5.96 (solid) against a highest control of 5.65 (dotted). (e) The 3 other blocks covering that cell in the star’s rest frame, combined, with zero at the *predicted* channel, which the frame transport moves between epochs; the nearest is 1.97 hours earlier than the discovery (2017 May 19), the farthest 3 days. A persisting carrier would have returned $T = 9.7$ (dashed) against 2.71 present, excluding it at 9.3σ . (f) The second event, CP-72 2713 at 344.269746 GHz: its discovery stack (black) and the same cell 2.05 hours later in a block 1.1 times deeper (blue), where a persisting carrier would have given $T = 6.2$ (dashed) against 3.60 measured. The inferred isotropic equivalent powers are $5.1 \times 10^{14} \text{ W}$ at 8.5 pc and $8.9 \times 10^{14} \text{ W}$ at 36.7 pc .

weighted sum, and at least 0.50 of the signal surviving removal of the strongest epoch. Controls are stacked identically and every clause evaluated at them, which measures the whole test’s false-alarm rate at 0.21 per cent against a 1 per cent ceiling fixed in advance.

The gain is $\sqrt{N_{\text{eff}}}$ and not $\sqrt{N}$, where $N_{\text{eff}} = \sigma_{\text{min}}^2 \sum_e \sigma_e^{-2}$ is the equivalent number of equal-depth epochs and the median N_{eff}/N is 0.65. What a stack realises is $\sqrt{N_{\text{eff}}}$ divided by the measured noise of its own stacked spectrum, and that noise follows the inverse-variance prediction to 0.992 in the median, with a per-stack scatter of 7 per cent and nine stacks in ten between 0.81 and 1.18, the sampling error of a noise scale measured over 127 channels. Against $\sqrt{N}$ the realised gain is 0.78, the shortfall being unequal epoch depth, so an epoch count alone misleads. HD161868 stacks 13 epochs for a gain of only 1.15: its N_{eff} is 1.7, so $\sqrt{N_{\text{eff}}}$ is 1.29, and its stacked spectrum came out 1.12 times as noisy as predicted.

The deepest stacked limit is Proxima Centauri’s, and it is a Class B limit. Proxima is covered by 66 blocks over 85 h spanning 6.3 yr, all of them coarse 15.625 MHz windows, so the stack constrains an unresolved spectral excess and discriminates no drift. 25 of those blocks belong to the primary census; 33 were searched afterwards from the same public archive and 8 come from the reserved hold-out, and all of them were observed over the same years as the census blocks themselves. Stacking the census blocks alone, which is the set every other limit here is

computed over, gives $\text{EIRP}_{90} = 2.2 \times 10^{13} \text{ W}$; adding the rest of the archive deepens that by 1.26 to $1.73 \times 10^{13} \text{ W}$, against $5.51 \times 10^{13} \text{ W}$ for the best single epoch on the same injection-calibrated convention. Its N_{eff} is only 7.9, so $\sqrt{N_{\text{eff}}} = 2.80$ of the measured 3.19 is what combining the epochs buys, the rest being this spectrum’s own noise, 0.88 of prediction over 126 channels. The limit holds for an emitter steady in the stellar frame to within one 15.625 MHz channel, 20.9 km s^{-1} peak to peak, with no inter-epoch drift above $7.8 \times 10^{-2} \text{ Hz s}^{-1}$. At 1.30 pc that depth is mostly geometry; the gain is what stacking bought.

2 groups pass the full criterion and both are known lines. They are β Pictoris’ CO $J = 1-0$ at $+3.09 \text{ km s}^{-1}$ over 8 epochs ($Z = 45.3$) and its $J = 2-1$ at $+3.55 \text{ km s}^{-1}$ ($Z = 12.1$), agreeing to 0.46 km s^{-1} or 0.73 of a CO(1-0) channel: the astrophysical positive control passing in the stack. The survey’s own $\pm 50 \text{ km s}^{-1}$ mask leaves 0 of them unattributed. The stack is also an independent test of the threshold crossings of §5.6: the 9 stars carrying them stack to $Z = 2.82-4.43$ in 8 cases, 0 exceed their own control ensembles and 0 passes the criterion. The exception, HD 285968, reaches $Z = 12.89$ on the foreground line used above as the registration standard and is rejected because its own controls reach 14.19: the emission is extended, as a cloud is and a transmitter is not.

The restriction travels with every number above, and its width is the channel. At CO(2-1) one 488 kHz chan-

TABLE 4

THE TWO EVENTS OF FIG. 6, THE UNATTRIBUTED CROSSINGS WHOSE STELLAR STATISTIC EXCEEDS ALL 512 CONTROL POSITIONS OF THEIR OWN WINDOW. REPEAT EPOCHS MEASURED ARE THE BLOCKS AT WHICH THE PREDICTED STELLAR-FRAME CELL WAS READ; THE INDENTED ROWS GIVE HOW MANY OF THOSE THE RELEASED CATALOGUE CARRIES AND HOW MANY BLOCKS IT LISTS AS COVERING THE FREQUENCY AT ALL, WHICH DIFFER BECAUSE THE ARCHIVAL EXTENSION IS DEPOSITED SEPARATELY. T_{rep} COMBINES THE REPEAT EPOCHS BY INVERSE VARIANCE: IT IS ONE CELL AND ONE TRIAL, SO IT MAY BE NEGATIVE, AND IT IS THE QUANTITY T_{pers} IS COMPARED AGAINST. THE INDENTED READINGS BENEATH IT ARE THE LARGEST SINGLE EPOCH AND THE LARGEST OVER EVERY DRIFT TRIAL, EACH A MAXIMUM AND SO POSITIVE ON AVERAGE WHATEVER IS THERE. THE EXCLUSION IS $(T_{\text{pers}} - T_{\text{rep}})/\sqrt{1 + (T_{\text{pers}}/T_{\text{rep}})^2}$, THE SIGNIFICANCE AT WHICH TWO READINGS OF ONE AMPLITUDE DISAGREE, AND THE LAST ROW IS THE FRACTION OF THE DISCOVERY AMPLITUDE S STILL RULED OUT AT THE TRIGGER, $(T_{\text{rep}} + 5)/T_{\text{pers}}$. THE RADIATED POWER QUOTED IS THAT IMPLIED BY THE FLUX IN THE PEAK CHANNEL; AN UNRESOLVED CARRIER LEAVES ONLY 0.48 OF ITS POWER THERE, SO THE POWER ACTUALLY RADIATED IS $\times 2.00\text{--}\times 2.36$ LARGER, A FACTOR ALREADY INSIDE EIRP₉₀ AND NOT INSIDE THIS ROW.

	61 Vir	CP-72 2713
Frequency (GHz)	346.313811	344.269746
Channel width (kHz)	488.28	488.28
Drift rate (Hz s ⁻¹)	+3182	-2056
$T_{\star}$	5.958	5.809
Highest control T	5.65	5.68
Flux density (mJy)	118.8 ± 19.9	11.3 ± 2.0
Peak-channel EIRP (W)	5.1 × 10 ¹⁴	8.9 × 10 ¹⁴
Repeat epochs measured	3	5
in the released catalogue	3	1
catalogue blocks at ν	4	1
Observed	2017 May 20	2022 Oct 2
Nearest covering epoch	2017 May 19	2022 Oct 2
separation (h)	-1.97	+2.05
covering epochs more than a day away	2	4
Longest baseline (d)	3	2
T expected if persistent, T_{pers}	9.70	6.30
T measured, matched drift, T_{rep}	+0.39	-0.82
largest single epoch	+1.65	+0.96
maximised over every drift	2.71	3.60
Exclusion, σ_{excl}	4.9 σ	4.8 σ
Amplitude still excluded at 5 σ	0.56 S	0.66 S
Disposition	not confirmed	attributed

nel spans 0.63 km s⁻¹, and that is the whole excursion a stacked carrier is allowed, peak to peak. A transmitter on a short-period giant planet, whose line-of-sight velocity swings by ± 30 km s⁻¹, crosses 94 of those channels peak to peak and so de-registers itself completely, against 3.0 channels of a 15.625 MHz window: the coarse stacks buy reflex tolerance with bandwidth. Drift divides the same way, a stack in 488 kHz channels being confined to 1.3×10^{-1} Hz s⁻¹ over the median inter-epoch span of 44 d. Each stacked limit therefore constrains only emitters stable in the stellar frame to within its own channel; the single-epoch limits of §6.3 apply generally.

6. DISCUSSION

6.1. What is different about a millimetre technosignature search

The most transferable result of this experiment is not the non-detection but a measurement of what limits a carrier search at these wavelengths. For classifying a crossing in the present archival data set, molecular-line confusion and sparse repeat coverage are more immediate limitations than thermal sensitivity. That is a statement about disposing of events and not about depth: the power a transmitter must radiate to be found here

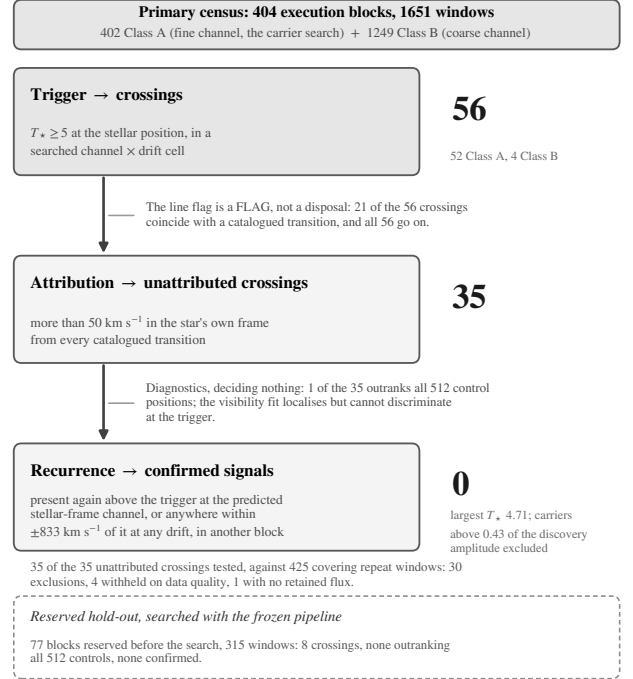


FIG. 7.— The detection logic, with the count surviving each level. Three levels and no more: a *crossing* is an event requiring investigation, an *unattributed crossing* is a target for follow-up, and a *confirmed signal* would be a detection. Coincidence with a catalogued transition is a flag and not a disposal, so all 56 crossings continue past it; the spatial rank and the visibility fit are diagnostics attached to the middle level and decide nothing. The reserved hold-out is drawn as a separate branch because a transmitter in it would matter whatever its calibration role. Every count is read from the ledger and the catalogue, so the figure, Table 8 and the text cannot disagree.

remains a median 1.6×10^{15} W per system.

Confusion, and not interference, sets the false-positive rate. Catalogued molecular transitions, almost all of them carbon monoxide, account for 21 of the 56 threshold crossings, and no crossing in this survey is attributed to interference (Appendix E). 10 of the 12 windows that reached the spatial stage are disc or foreground carbon monoxide, arising in 2 of the 69 debris-disc systems searched. The remedy that works is a velocity-space mask evaluated in the stellar frame, and it is not free: it removes 4.3 per cent of otherwise usable coverage.

Spatially extended emission is the second discriminant, and it has no analogue at centimetre wavelengths. A line filling the primary beam raises the star and the control annulus together instead of cancelling, so the spatial statistic loses power exactly where circumstellar gas is brightest. β Pictoris shows the benign face: the frozen pipeline recovers its circumstellar carbon monoxide blind, in two transitions, which is the positive control the experiment needed. HD 48370 shows the other. Its resolved disc emission lies near a band edge, and that one window contributes 59 per cent of every near-edge threshold exceedance in the survey: pooled, the near-edge rate is $\times 2.464$ the interior rate; without that window $\times 1.020$, and on a held-out half $\times 1.181$ ($p = 0.24$). What looked like a property of band edges is one disc, so no edge correction is applied to any limit.

The channelisation is inherited. 1249 of the 1651 windows have channels so coarse that every drift in the grid

TABLE 5

THE THRESHOLD CROSSINGS OF THE RESERVED HOLD-OUT, SEARCHED WITH THE SAME FROZEN PIPELINE AND DISPOSED OF BY THE SAME TESTS AS THE CENSUS. Δv_* IS THE OFFSET FROM THE NEAREST MASKED TRANSITION IN THE STAR’S OWN FRAME AND n_{rep} THE NUMBER OF COVERING REPEAT BLOCKS; T_{pers} AND T_{rep} ARE AS IN TABLE 8. THE ONE CROSSING THAT IS PRESENT AGAIN IS A CARBON MONOXIDE LINE.

Star	block	ν_{topo} (GHz)	T_* line	Δv_*	n_{rep}	T_{pers}	T_{rep}	disposition
HD 285968	Xdaef5a Xdd9b	230.5081	8.642 CO(2–1)	+8.7	4	11.6	+6.71	attributed; present again
HD 14055	X107376c X6126	330.7123	5.513 ^{13}CO (3–2)	+112.6	30	312.0	+0.56	unattributed; does not recur
TRAPPIST–1	Xd0e450 X259b	346.2454	5.326 SO(9 ₈ –8 ₇)	–306.6	5	11.1	–0.32	unattributed; does not recur
HD 172555	X107fa96 X1300f	344.6066	5.144 SO(8 ₈ –7 ₇)	+251.3	—	—	—	unattributed; no repeat coverage
HD 285968	Xdb4b9a Xa7c	229.9129	5.106 CO(2–1)	–768.0	4	11.2	+0.21	unattributed; does not recur
HR 1010	Xc6b674 X23f2	346.2301	5.044 SO(9 ₈ –8 ₇)	–238.7	7	25.9	–0.80	unattributed; does not recur
HD 10647	Xff99e1 X9cdf	345.7059	5.017 SO(2 ₃ –2 ₁)	+37.1	13	142.1	–0.93	attributed; does not recur
HD 170773	Xff294f X1fb	345.5597	5.001 SO(2 ₃ –2 ₁)	–113.8	4	59.3	+1.45	unattributed; does not recur

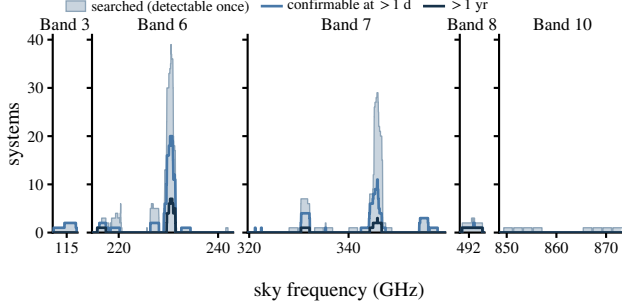


FIG. 8.— What could be searched against what could be confirmed. For the Class A experiment, against sky frequency: the number of systems toward which a carrier could have been detected once (grey), the number for which a second independent execution block more than a day later also covers that frequency (blue), and the same beyond a year (dark). One panel per populated ALMA band, widths in proportion to the frequency each contributes, as in Fig. 9.

stays inside one channel over a whole track; they measure an amplitude but cannot distinguish a drifting carrier from a stationary one, which is why the carrier search rests on 402 of them. The widths on offer here run from 15.3 kHz to 31.25 MHz, about 3.3 orders of magnitude; set beside the 2.8 Hz channels of centimetre-band carrier searches the range becomes 7 orders (Fig. 1(b)), and nothing in the search could choose them.

Repeat epochs turn a crossing into a testable claim, and an archive supplies them only incidentally. Let the *confirmation coverage* $C_{\text{conf}}(\nu)$ be the number of systems toward which two independent execution blocks more than a day apart both cover the sky frequency ν , so that a carrier found once there could have been sought again. It must be stated against frequency rather than as a fraction of the sample, because repeat visits re-tune and a second epoch is of no use at a frequency it does not cover. Its support is 23.3 GHz of union bandwidth, 49 per cent of the 47.7 GHz searched, falling to 8.5 GHz beyond a year, and at its best near 231 GHz it reaches 20 systems (Fig. 8). The searched extent is about twice the extent over which a persistent carrier could have been confirmed, and for close to half the sample the archive constrains an intermittent emitter hardly at all, whatever its power.

Finally, a threshold is not a sensitivity. A trigger level is a property of the noise, not of what a transmitter of a given power does to a pipeline: measured by injection and recovery through the unchanged selection, the power needed for ninety per cent recovery is 3.03 times a Class A window’s own trigger power and 4.17 times a Class B window’s.

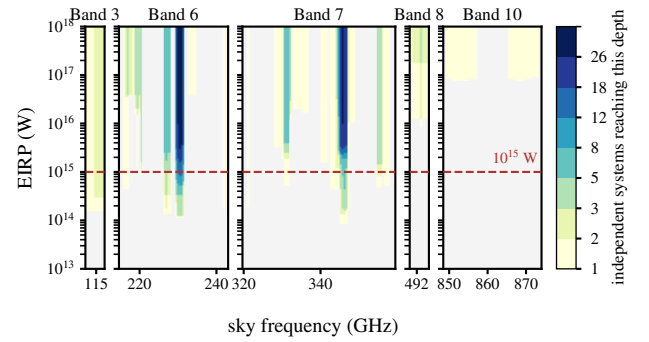


FIG. 9.— Where the experiment could have found something. Shading gives the number of independent systems that would have yielded a recovery at 90 per cent completeness for an unresolved carrier of the plotted EIRP at the plotted sky frequency; unshaded means none. One panel per populated ALMA band, with panel widths in proportion to the frequency each contributes: the 47.7 GHz Class A union is 26 disjoint islands across 114.0–873.1 GHz, which a continuous abscissa would draw as hairlines. The dashed line is 1×10^{15} W. Figure 2 shows where ALMA looked; this shows where the experiment was sensitive.

6.2. The search space this archive opens

The carrier search is 402 fine-channel windows toward 60 stellar systems, 47.7 GHz of union bandwidth; the 1249 coarse-channel windows are a supplementary spectral-excess experiment. No previous search has looked for technosignatures toward stars within 40 pc at any frequency above about 30 GHz, and the one earlier millimetre search worked in Band 3 toward far more distant stars (§2), so Band 3 is not searched toward nearby stars either. 20.6 GHz of the union here lies inside the 10 MHz–115 GHz interval Wright et al. (2018) adopt as the frequency axis of the search haystack.

Union bandwidth alone overstates what that buys: different frequencies were searched toward different numbers of stars and to very different limits. Figure 9 therefore gives the quantity a reader needs instead: how many independent systems would have yielded a recovery at ninety per cent completeness, against sky frequency and transmitter power. At an EIRP of 1×10^{15} W the covered frequency falls to 19.1 GHz, 40 per cent of the union, reaching 25 systems in all and 17 at the best-covered frequency near 231 GHz; with no limit on power that frequency reaches 39 of the 60 systems, and half the sample only above 3.4×10^{15} W.

Archival SETI buys sky and frequency cheaply and buys confirmation not at all. Across both classes this experiment reached 82 systems and 118.1 GHz of union bandwidth without an allocation, on data selected by others and therefore with the biases of §3.1.

For comparison with other programmes the conventional measure is the continuous-waveform transmitter figure of merit of Enriquez et al. (2017), $\text{CWTFM} = \zeta_{\text{AO}} \text{EIRP}_{\text{min}} / (N_{\star} \nu_{\text{rel}})$ with $\nu_{\text{rel}} = \Delta\nu / \nu_{\text{mid}}$, smaller being better. Those programmes quote EIRP_{min} at a bare detection threshold, so this one is put on the same footing: for the Class A experiment, with $N_{\star} = 60$ and $\nu_{\text{rel}} = 0.097$, it is 1.0×10^5 on the sample’s largest threshold power and 4.4×10^3 on the median system, against 0.85 for Enriquez et al. (2017). Margot et al. (2023) publish no CWTFM; evaluating the same expression on the three quantities they do give — 11 680 stars, 1.15–1.73 GHz and $\text{EIRP} > 1 \times 10^{13}$ W — returns 0.11. Carried through instead with the EIRP_{90} adopted everywhere else here, the same two become 3.2×10^5 and 1.3×10^4 . Nearly all of the difference is the product $N_{\star} \nu_{\text{rel}}$: this is a far smaller haystack than a wideband centimetre survey of thousands of stars, and the measure compares surveys rather than estimating an occurrence rate.

6.3. What this experiment excludes, and what it does not

Excluded. Consider a continuously transmitting, unresolved carrier that lies outside the molecular-line mask, inside the searched frequency range and inside the linear-drift grid, and that is present during one of the archival epochs. Such a carrier would have been found nine times in ten above the EIRP_{90} of §5.1, a median of 1.6×10^{15} W per system on its best window. None is present toward any of the 60 systems the carrier search covers.

Four further axes bound the excluded population as sharply as power does. *Frequency:* the coverage is sparse and inherited (Fig. 9) and the median system is covered over only 1.84 GHz. *Drift and acceleration:* the grid reaches a line-of-sight acceleration of $3.60\text{--}4.00 \text{ m s}^{-2}$, which contains every known planet of the sample but one (Fig. 3); an oscillator drifting faster than that lies outside it. *Persistence:* the carrier must be radiating during one of the archival observations. *Spectral width:* the statistic is built for emission confined to one 15.3–1953 kHz channel, and anything broad enough to resemble continuum is removed with the spectral baseline.

Not excluded. A transmitter absent during the archival observation, or on for much less than the length of one. The limits scale inversely with the duty cycle (§5.1), so a carrier radiating a tenth of the time is constrained only at $10\times$ the powers quoted here, and because the intermittency trials carry no drift the joint drifting and intermittent case is not measured at all. Emission inside the $\pm 50 \text{ km s}^{-1}$ molecular-line mask, at any strength, which costs 5.04 GHz of union bandwidth. Drift outside the searched grid, or drift curved enough to leave a channel within the track. Signals broad enough to be removed with the continuum baseline. Anything below a window’s own completeness. And any emission from the 17 566 stars within 40 pc that ALMA has not observed, which is almost all of them.

An illustrative engineering benchmark. A 12 m dish radiating 1 MW at 230 GHz with an aperture efficiency of 0.7 has a directive gain $G = \eta(\pi D \nu / c)^2 = 5.9 \times 10^8$ and so an EIRP of 5.9×10^{14} W, a depth this survey reaches for 16 of the 60 Class A systems. The same gain fixes how narrow the beam is, and detection requires the Earth to lie in-

side it: a solid angle $\Omega = 4\pi/G = 2.1 \times 10^{-8}$ sr, about 34 arcsec across, illuminating a fraction $\Omega/4\pi = 1.7 \times 10^{-9}$ of the sky at a time. The example shows that EIRP is not transmitter power — the same megawatt radiated isotropically would be 10^6 W — so these limits apply to beamed transmitters whose beam contains the Solar System and say almost nothing about omnidirectional beacons.

Two blind spots. The search is Stokes I only, so the discrimination that polarisation would offer against unpolarised astrophysical emission is unavailable; ALMA’s feeds are linear, and the circularity of a circularly polarised carrier lies in the cross-hands, which this analysis does not extract. And the frequency coverage is the archive’s and not a choice: none of the “magic frequency” arguments that have guided centimetre-band searches falls in an ALMA band, so a transmitter deliberately placed on one lies outside this experiment by construction.

6.4. What a dedicated ALMA technosignature experiment should do differently

A purpose-built experiment on the same telescope would be far stronger, and the changes it needs follow from the limitations above. *Channelise finely everywhere:* the 1249 coarse-channel windows measure an amplitude but cannot distinguish a drifting carrier from a stationary one. *Choose the targets and the frequencies:* an archive weighted toward dusty discs is the worst case for molecular-line confusion, the M dwarfs that dominate the nearby habitable-zone planet census are the least covered, and windows placed deliberately away from the bright transitions would recover most of the 5.04 GHz the mask costs — all three are decisions a proposal can simply reverse. *Repeat the epochs on purpose, at the same tuning:* two or three visits separated by days and by years turn a crossing into a testable claim, and the separation is a scheduling constraint rather than extra sensitivity; Fig. 8 is what its absence costs. *Retain the visibilities:* rotating the phase centre to the star leaves a genuine point source entirely in the real part of the visibility. That asks whether the data contain a source *where the star is*, which is a stronger question than any rank against neighbouring positions in an image.

7. CONCLUSIONS

1. The frequency range is what makes this experiment new: no published radio technosignature search above about 30 GHz has reached a star within 40 pc (§2), and an archive assembled for other science is at present the only way to reach one.
2. The primary experiment is 402 fine-channel windows toward 60 stellar systems, covering 47.7 GHz of union bandwidth in disconnected intervals, searched for an unresolved carrier whose frequency drifts. A supplementary coarse-channel search for an unresolved spectral excess adds 1249 windows toward 81 systems and 90.5 GHz of union bandwidth, with no drift discrimination. Both come from one uniform reduction of 404 public execution blocks, 4.1 TB of raw data, and between them they span 90–873 GHz.
3. Sensitivity is a measured completeness, not a threshold. Carriers injected into the real calibrated visi-

bilities are recovered through the unchanged selection — the trigger, and localisation at the star — and EIRP_{90} is the power at which nine in ten come back. The median system’s best window reaches 1.6×10^{15} W, with a spread across systems of 6.8×10^{13} – 3.7×10^{16} W. The rank and the visibility fit decide nothing, so neither is charged to that limit. One bias runs one way only and is not inside the quoted calibration: an injected carrier suffers no atmospheric decorrelation where a real one would, so every limit here is optimistic by up to 20 per cent.

4. All 56 threshold crossings are reported with their dispositions, four of them in the one execution block that fails the data-quality criterion of Appendix D.2 and reported apart from the 52 in quality-passing data; of those, 21 coincide with a catalogued molecular transition, almost all of them carbon monoxide — including the β Pictoris disc, which the search recovered without being told where to look — and 31 do not.
5. Every unattributed crossing was tested for recurrence in the star’s own frame, in each further execution block covering the event frequency and over the whole of each such window at every drift searched, because a carrier on an accelerating platform need not return to one channel. None is present again: the largest statistic at any predicted cell is 4.71 against $T_{\text{trig}} = 5$, and the exceedances found anywhere in those windows are no more than chance predicts for trials of that size. Where a discovery amplitude survives, the repeat coverage rules out a carrier of that amplitude in 30 cases, and one of half that amplitude or less in 18 of them; taken instead as a disagreement between two measurements of the same amplitude, discovery and repeat differ by 3.4 to 5.6σ , median 4.9σ , which is as much as the pair can say when the discovery amplitude is itself a five-sigma measurement. Nor did the one ahead of all 512 control positions of its own field return. What these tests exclude is a carrier that holds its frequency inside the window searched and radiates through both observations, and not a persistent transmitter of any kind.
6. What this survey could search and what it could confirm are different extents. The carrier search covered 60 systems, of which only 30 — carrying 49 per cent of the searched union bandwidth — were observed again at the same frequency more than a day apart, and 11 beyond a year. Across both classes, 44 of the 82 systems have no second block covering a frequency the first one covered, and toward those the null reported here says nothing at any strength.
7. Combining each surveyed star’s epochs in its own rest frame deepens the flux limit by a median factor 1.52. For Proxima Centauri, where $\sqrt{N_{\text{eff}}} = 2.80$ is what combining the epochs buys, the stacked limit measured by injection into the stack reaches 2.2×10^{13} W over the primary census alone and 1.73×10^{13} W once the rest of the archive is added, in 15.625 MHz channels — a Class B limit on an unresolved excess, carrying no drift discrimination — for an emitter steady in the stellar frame to within one channel, 20.9 km s^{-1} peak to peak, across 6.3 yr.

8. These results constrain unresolved carriers only within the domain actually searched and only toward this archive-selected sample (§6.3). What transfers beyond it is the obstacle: a channelisation chosen by other observers for other purposes, and not collecting area, is what decides which kind of search is possible at all.

AUTHOR CONTRIBUTIONS

G.J.W. conceived the survey, defined the target sample and the selection criteria, and wrote the manuscript. R.D. built and operated the archival retrieval, calibration and search pipeline, implemented the spatial control statistic and the injection–recovery campaigns, and produced the released catalogue and its regeneration scripts. Both authors designed the vetting procedure, chose the masked species, carried out the statistical validation reported in §4.1 and Appendix D, and agreed the dispositions of Table 8.

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COMPETING INTERESTS

The authors declare no competing interests. VBRL Holdings is a privately held company through which R.D. conducts independent research, named here solely as his affiliation; it had no role in the design of the survey, the selection of targets, the analysis, the decision to publish or the content of this paper, and provided no funding for the work.

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This paper uses ALMA data from the public ALMA Science Archive. The 404 primary-census execution blocks searched span 2013 October 6 to 2025 June 11 and belong to 65 ALMA proposals, whose codes, in the standard ADS/JAO.ALMA#*code* form, are recorded block by block in the data release together with each block’s epoch. ALMA is a partnership of ESO (representing its member states), NSF (USA) and NINS (Japan), together with NRC (Canada), NSTC and ASIAA (Taiwan), and KASI (Republic of Korea), in cooperation with the Republic of Chile. The Joint ALMA Observatory is operated by ESO, AUI/NRAO and NAOJ. This work has made use of data from the European Space Agency (ESA) mission *Gaia* (Gaia Collaboration et al. 2023), processed by the *Gaia* Data Processing and Analysis Consortium (DPAC). This research has made use of the SIMBAD database (CDS, Strasbourg, France), of the NASA Exoplanet Archive (Akeson et al. 2013), and of the Splatalogue spectral-line database (Remijan et al. 2007).

DATA AVAILABILITY

The reference of record is a single permanent Zenodo deposit, cited by its DOI. It carries every number in this paper and the code that produced them, and contains: the 89-star target catalogue with its selection function; one row for each of the 1651 searched windows,

carrying its execution-block identifier, frequency limits, native channel width, trial-drift grid, trigger statistic at the stellar position, control statistics of the 512-position ring and the window’s EIRP₉₀; the ALMA proposal code and epoch of every execution block, so that each of those rows resolves to the programme that took it; all 56 threshold crossings with their stellar-frame frequencies, line classifications, recurrence measurements and dispositions; the search statistic and the molecular-line list as files in their own right; the injection–recovery records; the block-by-block archival verdicts and the ALMA delivery record for every block in scope that could not be searched; the diagnostic figures the printed paper refers to but does not reproduce, among them the per-system coverage waterfall and the selection funnel (`coverage_waterfall_full.pdf`, `selection_funnel.pdf`); the illustrative table of sample systems within 15 pc with a confirmed planet; and the full pipeline with the commit that produced the deposited products. Two properties of the deposit are measured rather than claimed: 146 generated products regenerate byte-identically from the frozen inputs, and 41 headline quantities are re-derived from the released catalogue alone, without the pipeline.

Three of the choices that could create or remove an unattributed crossing carry an identifier that can be checked independently of the deposit. The calibration hold-out rule of Appendix B.1 and the block list its first guard depends on were committed to the project repository as `c75069040eab52ec1336376a4dae62114bb281a3` on 2026 September 14 at 07:10 UTC, 3 days before the first reserved block was searched. The molecular-line list is a single dated query of a public line catalogue (2026-10-06) deposited as a file, so it is reproducible rather than described; it post-dates the fixing of the disposition criteria on 2026 September 11 (`f5f52e7804f1baad99515ea2d46ddd7e4e4cd546`), and so does every species in it, which is why the same rule is applied to all of them and no species is withheld from one coincidence and allowed in another (Appendix C). The repository at <https://github.com/Tilanthi/SETI> is a working mirror and may move on.

SOFTWARE

The pipeline depends on CASA 6.7.6 (casatools/casatasks 6.7.6-14; CASA Team et al. 2022), Python 3 with astropy (Astropy Collaboration et al. 2022), numpy (Harris et al. 2020) and scipy (Virtanen et al. 2020), and the ALMA Science Archive and SIMBAD TAP interfaces through astroquery (Ginsburg et al. 2019); all are free and open-source, with exact versions recorded in the frozen release.

APPENDIX

SPECTRAL RESPONSE, FREQUENCY CONVENTIONS AND THE PRIMARY-BEAM RESPONSE

Checking a threshold in this paper needs three things: the frame every frequency is quoted in, the instrumental response that separates a channel threshold from a transmitter power, and the primary-beam factor that turns a measured noise into a flux at the star. The per-window catalogue, `catalogue/windows.csv` in the deposit, carries the columns each of them uses.

Frequency frames, linewidth and smearing

Every frequency axis here — searched spectra, tabulated ranges and crossing frequencies alike — is *topocentric* as delivered by the ALMA correlator, and the search itself applies no velocity-frame shift. Velocities enter at the vetting stage, where the line mask is evaluated in each star’s own rest frame: it carries every crossing frequency to the barycentre at its own block’s epoch and then to the stellar frame before comparing it with a transition (Appendix C). The residual error is then the catalogue radial-velocity uncertainty alone, a few km s^{-1} within 40 pc; second-order relativistic terms at the 50 km s^{-1} scale are $\lesssim 5 \text{ kHz}$ at 350 GHz, below the finest native channel searched. The stacking is drift-aligned and inverse-variance weighted: coherent in frequency trajectory, not in electric-field phase.

The native channel width is also the linewidth convention for every EIRP threshold. The statistic is the single-channel flux density after drift correction, so $\text{EIRP}_{\text{trig}}$ is the minimum *total* power of a channel-confined carrier, flat in the assumed linewidth below the channel and never a spectral power density; flatness is exact only for a centred tone, and a boundary-straddling one divides its power between two channels, which is the loss the injections carry and one reason EIRP values are quoted to two significant figures. The Hz-resolution surveys’ limits are total powers in the same sense, each the best case of its own channelisation, which makes the axes of Fig. 1(a) commensurable. One number fixes the scale, illustratively and not achievably: rechannelising the median Class A window’s 488 kHz to the 2.79 Hz of a dedicated narrowband search would divide the threshold by 418, and no reprocessing recovers resolution the correlator never recorded.

Channel widths come from the measurement sets’ own `CHAN_WIDTH`; MS v2 defines no “EFFECTIVE_BANDWIDTH” column, so the quoted channelisations are separations and are not claimed to be noise bandwidths.

ALMA applies online Hanning smoothing by default (Cortes et al. 2024), so the spectral response is ~ 2 channels wide and neighbouring channels are correlated. For the 0.25/0.5/0.25 kernel the coefficients are $\rho_1 = 2/3 = 0.667$ and $\rho_2 = 0.167$, with $\sigma_{\text{smoothed}}/\sigma_{\text{raw}} = 0.612$: the correlation is not confined to adjacent channels, the lag-two term being a sixth. The peak channel holds 0.50 of an unresolved carrier’s flux at a channel centre and 0.42 on a boundary, a median 0.48 over sub-channel phase, so a nominal threshold is optimistic by $\times 2.00$ – $\times 2.36$; the released catalogue carries the nominal threshold and both corrected values for every window.

The injected carriers are deposited with the correlator’s own channel response, at a sub-channel phase drawn afresh for each tone, so the smoothing loss sits inside the measured completeness rather than being applied to it. The deposited profile is the tone divided between the two channels it straddles and then passed through the three-point kernel, which leaves 0.500 of the flux in the peak channel for a carrier at a channel centre and 0.375 for one on a boundary, against 0.4244 for the exact transform of the lag window; averaged over phase the deposition therefore falls 8 per cent short of the ideal response, and the completeness is conservative by that much. Where in

a channel a carrier was placed makes no difference to its recovered signal-to-noise: the medians over 3673 tones agree across the whole range of sub-channel phase to 1 per cent, because a carrier at a typical searched drift rate crosses about 11 channels during one observation and so samples the whole response. Since the loss is carried end to end it must not be applied to the result as well: at the ninety per cent point of the recovery curve the peak channel holds $1.93\times$ the nominal trigger flux, which is that factor of about two, already spent.

Two measurements fix the smoothing state. The retained per-integration residuals of the second-stage local null have a lag-one spectral autocorrelation of 0.65 against the Hanning prediction 0.667, so the correlation is instrumental and those spectra are smoothed and unaveraged; and the archive's reported effective spectral resolution is exactly twice the channel separation in 1505 of the 1651 windows, in both correlator modes. The other 146 report a smaller ratio, the signature of online channel averaging, where the peak-channel loss is smaller; every released window has a reported resolution, so the blanket factor is right for the great majority of both classes. *The correlation model is truncated at lag two*: the baseline step, a per-integration median over up to 65 channels, could imprint longer-range correlation that the released products retain no residuals to measure, which would make the independent-trials count of Appendix D an over-estimate.

Transmitters wider than one channel. The baseline filter (§4.1) is flat for features narrower than roughly half its effective width W and falls to zero above W , so the survey's sensitivity is to channel-confined morphologies and W is the linewidth ceiling: 65–74 channels over the fine class and 30–59 over the coarse one, that is ≈ 32 MHz at 488 kHz and ≈ 0.9 GHz at 15.62 MHz. Within that ceiling a top-hat transmitter spanning N_{ch} native channels has per-channel flux the total divided by N_{ch} at unchanged noise, so its threshold is N_{ch} times the single-channel one, which is the rescaling the main text applies to signals broader than one channel.

Intra-integration smearing. A drift smeared by Δ channels within one integration keeps $\eta_{\text{smear}} = \text{sinc}(\pi\Delta/2)$ of the peak amplitude, so the effective threshold is $\text{EIRP}_{\text{trig}}/\eta_{\text{smear}}$. The excursion can be evaluated wherever the release carries the integration count, which is 451 of the 1651 windows; over those it reaches at most 1.10 channels. Only 9 of them have $\eta_{\text{smear}} < 0.99$ — HIP 10679 Band 6 (0.574, $\times 1.74$), β Pic Band 6 (0.574, $\times 1.74$), HD 172555 Band 6 (0.574, $\times 1.74$), HIP 10679 Band 6 (0.606, $\times 1.65$), HD 172555 Band 6 (0.606, $\times 1.65$), η Crv Band 8 (0.865, $\times 1.16$), η Crv Band 7 (0.932, $\times 1.07$), HD 48370 Band 6 (0.979, $\times 1.02$) and CP–72 2713 Band 6 (0.979, $\times 1.02$) — and the other 442 agree with their nominal thresholds to better than one per cent. Compounding the worst smearing penalty with the worst Hanning factor gives $\times 4.6$ for those few, a product no released column carries.

The primary-beam response, and the two columns that let a reader recompute it

The sensitivity quoted for a window is $S_{\text{min}} = 5\sigma/A$, where A is the primary-beam response at the star's offset from the phase centre: the noise σ is measured in apparent flux, the limit is quoted in true flux, and the

two differ by A . That offset is not a nuisance. Archival fields are pointed at a catalogue position, often decades old, while these are high-proper-motion stars, so the star can sit well off axis: over the 1651 released windows the offset has a median of $0.089''$, a 99th percentile of $7.43''$ and a maximum of $8.13''$, and A runs from 0.999984 in the median down to 0.5522, below 0.99 on 91 windows and below 0.9 on 34. *None* falls below one half, and every one of the low-response windows is a physically explicable J2000 pointing on a fast-moving star rather than an error. A window is kept only where the response is at least 0.5, a correction of at most $\times 2$, since at low response a small error in the beam model becomes a large error in inferred flux. The offsets are bimodal and the floor sits in the gap: the last retained window has response 0.55 at $0.46\theta_{\text{PB}}$, the first window beyond it 0.35 at $0.62\theta_{\text{PB}}$, and nothing lies between, so the rule binds prospectively and excludes none of the 1651 released windows. Beyond it lie the 12 ϵ Eri Band 6 windows, 8 of them needing a correction of up to $\times 3.5$ — past the first sidelobe transition, where the Gaussian form fails, and 4 needing none. The stratum is *withheld* entire rather than window by window, so none of the 12 enters a limit or the released catalogue. The flux-scale comparison of §4 is a separate continuum measurement on the same star and uses none of them: it is a block mean over the 4 compact-array executions that place ϵ Eri at the pointing centre, 0.22 h in Band 6, where no beam correction enters at all.

The correction is applied exactly once. The offset is computed for every measurement set from its own `FIELD::PHASE_DIR` and the propagated stellar position, so no window carries a zero-by-assumption offset, and it enters in one place only: $S_{\text{min}} = 5\sigma/A$ closes on all 1651 released rows to $8.7\text{e-}05$ in relative terms. The count is in two parts, because on 10 rows whose products are not retained A is back-solved from $5\sigma/S_{\text{min}}$ and flagged as such, so there the closure is arithmetic; the audit is a test on the other 1641 rows, where A comes from the search's own stored geometry and closes to $8.7\text{e-}05$. Nothing else needs correcting: the statistic of Eq. 1, the control ranks and the attribution are *ratios* formed inside one window, in which A divides out identically.

The response applied is `pb_atten`: a Gaussian of full width $1.22\lambda/D$ — an Airy first-null coefficient — at each window's *own* dish diameter, reproduced over all 1651 released rows to 2.6×10^{-5} against 7.7×10^{-2} for the same Gaussian forced to $D = 12$ m. The deposited `theta_pb_arcsec` is that width at $D = 12$ m for every row, including the 1054 taken on the $1.7\times$ smaller compact-array dishes, so A must be read from `pb_atten` and cannot be recomputed from it.

The beamwidth convention is a bounded systematic, and the release carries both conventions so that a reader need not take ours. ALMA's primary-beam full width is $1.13\lambda/D$, narrower than the first-null coefficient, so it attenuates more and makes every limit *shallower*: by nothing at all in the median window, by more than one per cent in 40 of 1651 and by more than ten in 4, the largest being $+10.3$ per cent — those few all LP 736-15 at $8.1''$ off axis. The lowest response in the release moves 0.5522 to 0.5005, still clear of the 0.5 retention floor but by a margin small enough to be worth saying. `pb_atten_fwhm` in the deposit carries that column window by window, with each window's own θ_{PB} beside it.

Changing the convention changes no disposition, because A divides out of every ratio formed inside a window, and the published identity $S_{\min} = 5\sigma/A$ is left closing on the response that formed it. The remaining model uncertainty is also bounded by measurement: CASA’s blocked-Airy response is better physics than either Gaussian, the largest ratio to it over the whole release is 1.0745, and the counts of systems reaching 10^{15} W are identical under any of the three.

The star’s apparent direction includes the annual parallax, and the amplitude a compact source retains when that is left out of the assumed phase centre is measured rather than modelled. The displacement decides whether the loss can be treated as a scalar at all: a small fraction of a synthesised beam attenuates a compact source and does nothing else, while a displacement approaching a beam puts the phase wrong on the longest baselines. Re-extracting the 8 windows of 2 blocks furthest out at the full apparent position — proper motion, parallax, and radial velocity where catalogued — leaves each window’s own noise unchanged to 0.20 per cent and alters no disposition, but it cannot test the amplitude factor, *because the factor describes a source and those windows hold none.* The statistic at the star is a maximum over a plane of drift trajectories, so moving the extraction point by two thirds of a beam draws a nearly independent noise realisation: the ratio scatters from 0.78 to 1.15 about a median of 0.98, nowhere near the 0.49 the model predicts for a source. So a source was put there. A carrier injected at the apparent position and recovered from the *same* visibilities at both positions returns a ratio that is the normalised dirty beam at the omitted displacement, measured on that block’s own baselines with no model in it: at 0.25 of a beam τ Cet returns 0.92 where the model gives 0.89, and at 0.67 γ Lep returns 0.58 against 0.49, 3 and 20 per cent above the prediction. That it exceeds the model at the larger displacement is the search’s doing: maximising over a plane of trajectories lets a partly decohered carrier still find a cell, so the pipeline retains more than the beam amplitude alone and a limit corrected by the model is shallower than it needs to be.

The injection campaign is blind to A by construction, and the bias that follows is exactly zero. The injector deposits an unattenuated amplitude at the star’s own direction cosines and refers the ladder to the *apparent* 5σ on all 40 scored units, so an injected apparent amplitude recovered nine times in ten implies a true flux larger by $1/A$ while P_{trig} is the power of a true $5\sigma/A$ and A cancels identically; 4 of the units have apparent and corrected $S_{\min}$ differing by more than a part in a thousand, so the cancellation is tested rather than vacuous. Being self-consistent in apparent flux, the campaign cannot falsify the beam model; and it reaches only $A \geq 0.9102$, whereas 75 windows on 5 stars lie further off axis, down to $A = 0.552$, so a limit quoted for one of those rests on an extrapolated curve; their 6 crossings are ratios and unaffected.

Four rows carry a catalogued noise that no retained product reproduces, and they are declared rather than reconciled: 4 windows of HD 285968 in block A002_Xd98580_X3bf6, whose catalogued σ is 0.88–1.32 per cent *higher* than the surviving product’s, so the limits there are weaker and not stronger. With the 10 windows whose products are not retained at all the declared prove-

nance gap is 14 windows, and the attenuation columns are unaffected.

THE SPATIAL RANK, AND WHAT IT CAN SUPPORT

The spatial rank asks whether an excess sits at the star or anywhere else in the field. It is a diagnostic, no disposition rests on it, and this appendix is the measurement that settles why: The star-versus-control rank is an empirically useful way of deciding which crossing to examine first, but the stellar and control positions are not demonstrably exchangeable under the null. The rank therefore has no calibrated per-window false-alarm probability and is not used as evidence for detection.

The star lies at the phase centre, inside the control annulus, and the asymmetry is real. *It is not a primary-beam effect.* Spectra are extracted from phase-rotated visibilities in apparent flux and are never corrected for the primary-beam response (Appendix A), so the familiar rise of image noise with distance from the pointing centre cannot operate here; and were it operating, it would make the control probes noisier than the star and place the stellar rank *above* one half, the opposite of what is measured. **No mechanism for the displacement is established, and we prefer to say so.** What the control ensemble does show is a weak rise of its own level with radius, and the obvious repair is to remove it: each probe is standardised within its own radius bin, the radii being known from the fixed draw, with no free parameter.

Applied out of sample, the correction does not repair the statistic. Frozen — probe radii, bin edges and standardisation alike — and applied unmodified to 778 windows that played no part in its construction, it moves the median stellar rank from 0.409 to 0.407 against the exchangeable one half, further from it, in both out-of-sample partitions separately. Rescoring the whole survey with the gradient removed moves the list of crossings out-ranking all their controls from 12 windows to 11, losing 2 and gaining 1, and leaves the survey median rank slightly further from one half than before.

The radial gradient is real, but removing it leaves the median no closer to one half, so the residual is not radial and neither normalisation yields a calibrated tail probability. **We therefore keep the frozen rank, read it only as an indication of which crossing to examine first, and rest every disposition on physical evidence or on recurrence.** The radial correction, the region-maximum audit and the pseudo-star tests each leave every disposition unchanged, and because the rank gates nothing, none of this enters a sensitivity limit.

One mechanism can be excluded by measurement, and it is the one that would matter most. Equation 1 divides every position by a single noise scale shared by the star and all its probes. A star bright enough to carry multiplicative calibration residuals of its own would then have a noise the shared scale does not describe, and its statistic would be inflated with nothing in the search to reveal it. The ratio of the two noises is measurable from the released control vectors alone: because the star and each probe are divided by the same scale and maximised over the same channel $\times$ drift grid, the stellar statistic divided by the median of the 512 control statistics *is* that ratio, and it is one when the two noises agree. Over the 1595 windows carrying no threshold crossing its median

TABLE 6

THE BLOCK AND WINDOW LEDGER. EVERY SET OF EXECUTION BLOCKS USED ANYWHERE IN THIS PAPER, WITH THE RESULTS IT ENTERS; THE ROWS SUM, AND EACH INDENTED GROUP SUMS TO THE ROW ABOVE IT. THE PRIMARY CENSUS IS THE SURVEY: UNLESS A STATEMENT NAMES ONE OF THE OTHER SETS, IT IS COMPUTED OVER THAT ROW. HOURS ARE WINDOW-HOURS: ON-SOURCE TIME SUMMED OVER SPECTRAL WINDOWS, SO A BLOCK DELIVERING SEVERAL WINDOWS AT ONCE CONTRIBUTES ITS ON-SOURCE TIME ONCE PER WINDOW. THE CLASSES THEREFORE PARTITION THE HOURS AS THEY PARTITION THE WINDOWS. THE TELESCOPE ON-SOURCE TIME BEHIND THE PRIMARY CENSUS IS 280 h, OF WHICH 152 h CARRIES AT LEAST ONE CLASS A WINDOW AND 276 h AT LEAST ONE CLASS B WINDOW — FIGURES THAT OVERLAP AND MUST NOT BE ADDED. CONFIRMED PLANET COUNTS, HERE AND IN THE DEPOSITED TARGET LIST, ARE TAKEN FROM THE NASA EXOPLANET ARCHIVE **PSCOMP** TABLE (AKESON ET AL. 2013), ACCESSED 2026 OCTOBER 4.

Set of execution blocks	Blocks	Windows	Stars	Systems	Hours	Results it enters
Archival scope: blocks the query exposes	656	—	—	—	—	defines the scope; the sum of the next two rows
... processed with the frozen pipeline	479	—	—	—	—	the second panel below
... not processed at the freeze	177	—	—	—	—	the third panel below
Processed with the frozen pipeline ^c	484	—	—	—	—	the sum of the three rows below
Primary census	404	1651	89	82	1094	every limit, the crossing list, every chance expectation
Class A, fine channels ^a	—	402	65	60	227	the drifting-carrier experiment
Class B, coarse channels ^a	—	1249	88	81	867	the spectral-excess experiment
Reserved hold-out	77	315	36	—	214	the rank displacement, the false-alarm tail factor, the radial control profile
No window survived the quality cut	3	0	0	0	0	nothing
Not processed at the freeze: what became of each	177	—	—	—	—	the sum of the five fates below
Calibrated and searched	144	—	—	—	—	the sum of the four rows below
Archival extension , reported	115	463	16	16	269	recurrence, and the null on data the frozen pipeline had not seen
Searched after the extension was frozen	8	32	—	—	—	nothing; later than the reported extension
Withheld: primary-beam form invalid ^d	8	32	—	—	—	nothing; withheld
No window reached the export	13	0	—	—	—	nothing
No calibration in the ALMA delivery	18	—	—	—	—	nothing; the archival gap
Calibration failed	11	—	—	—	—	nothing; the archival gap
Not attempted	3	—	—	—	—	nothing; the archival gap
Working disk insufficient	1	—	—	—	—	nothing; the archival gap
Beyond 40 pc: out-of-sample control	47	198	22	—	88	the external null; no science count
External calibration sample ^b	326	1322	56	—	—	the rank calibration and the positive control; no science count

^aThe two classes partition the census windows, not its blocks: 252 blocks carry a Class A window and 389 a Class B one, and most carry both, so the hours of the two rows partition the windows and not the observing time.

^bNot a subset of the archival scope: it includes blocks toward stars outside the 40 pc work list, so its blocks cannot be placed inside this ledger.

^cThis panel and the scope panel above overlap rather than add: 479 of these 484 blocks are the in-scope row above, and the other 5 carry no member observing unit in the archive metadata, so the scope query does not expose them. 3 of the 5 are in the primary census.

^dEvery one is an ϵ Eri Band 6 block, where the Gaussian primary-beam form is not valid; the same rule withholds 12 windows from the primary census.

is 1.001 (star-clustered 95 per cent 0.994–1.004), and it does not climb with the brightness of the field, which is what a multiplicative error would scale with: 1.002 in the faintest fifth of control rings against 0.999 in the brightest, and 1.09 over the 8 windows whose ring exceeds 14σ and is therefore set by resolved emission rather than noise. That the ratio *can* rise is shown by the 56 windows that do carry a crossing, where it is 1.18. Part of the reason it does not rise otherwise is structural: a gain error multiplying a continuum raises every channel of an integration by the same factor, and the per-integration median along frequency removes exactly that. **A stellar-position noise denominator would therefore move no statistic in this survey.**

The reserved hold-out, and what it calibrates

The statistic and the mask were settled on the searched sample and cannot also calibrate themselves, so a hold-out was reserved by a rule fixed and recorded before any reserved block was searched: a block is held out if and only if $\text{sha256}(\text{UID})_{[8]} \bmod 5 = 0$, subject to two guards that use identifiers alone — a block on the pre-registered list deposited with this paper stays in the survey, and no system loses every block — so that the assignment can be recomputed from the UIDs and that list alone. The

statistic, the trigger and the recurrence criterion were all fixed before the rule itself, and the first reserved block was searched 3 days later, so every diagnostic built afterwards is incapable of promoting a window. Of 484 blocks processed, 77 are reserved (15.9 per cent), giving 315 windows toward 36 stars with no star and no system lost from the headline sample. Three quantities are calibrated on those blocks and on nothing else.

First, the rank displacement. Out of sample the stellar rank has median 0.405 where one half is expected, with a block-clustered 95 per cent interval of 0.341–0.458 over 315 windows, and 14 per cent of ranks below 0.1 against 6 per cent above 0.9, so the distribution is displaced rather than growing a tail. The displacement runs the way that matters: a rank below one half places the star high, and the star in fact outranks all 512 probes in 14 of the survey’s windows against 3.2 if the positions were interchangeable. The direction therefore inflates star-first outcomes rather than suppressing them, which is why the chance expectations of Appendix D are calibrated on the measured rate. Clustering removes neither the shift nor its size: 99 of 149 blocks have a mean rank below 0.5, and a star-clustered bootstrap places the median at 0.440 (95 per cent 0.42–0.46). The searched sample leans the same way but by less (median 0.49, star-clustered 95

per cent 0.47–0.51), and the two intervals do not overlap, so the displacement is a property of these reserved blocks and is not reproduced in sample. Either way it runs in the direction that inflates star-first outcomes, which is the quantitative reason the rank is read as a diagnostic and never as a significance.

Second, the false-alarm tail factor — the measured rate at which a no-signal position ranks first, divided by the rate exchangeability would give. It is 1.2 on 5040 trials, with a bootstrap clustered over the 77 reserved blocks giving 0.6–1.9. It is a statement about the rank and not about the trigger, so it is reported as a diagnostic; the quantity the chance expectations need is the exceedance rate at the stellar position, measured on the same blocks in Appendix D.

Third, the radial behaviour of the control ensemble: the standardised level runs +0.12, +0.10, +0.12, +0.16 across the annulus, a mean per-window slope of $+0.067 \pm 0.033$, small and of the wrong sign to account for the displacement.

The *external calibration sample* must not be confused with the hold-out, and it is not an out-of-sample set. Its 326 execution blocks were reserved to calibrate the rank and to supply repeats the hold-out does not have, and 263 of them are primary-census blocks and 63 hold-out blocks, with 0 outside both: it enters no count of the science sample because its blocks are already counted there, not because it is external. Over its 1322 windows the frozen pipeline returns 7 crossings outranking all their controls — 4 of them β Pictoris CO — and 3 unattributed against 3.7 expected at the measured tail rate.

One crossing through the whole chain

CP–72 2713 is the clearest worked example, because every step returns a verdict rather than a shrug. In Band 7 the crossing reaches $T_\star = 5.809$ in channel 35 of its window; it is unattributed, with the one nearby transition and its provenance discussed in Appendix C; both diagnostics favour it, in that it outranks every one of the 512 control positions and the drift-following visibility fit puts the emission at the stellar position, $\text{Re}/\sigma = 5.1027$ against a control-position maximum of 1.8222; and neither of those is a promotion (§5.5).

What disposes of it is the repeat coverage. One of its 5 covering blocks matches the discovery in tuning, channel width, channel count, integration count and on-source time, registers to 0.18 of a channel, and is the deeper of the two: a persistent emitter at the discovery amplitude would have returned $T_{\text{pers}} = 6.21$ there, and that block alone puts the pair 4.6σ apart. Combining all 5 gives $T_{\text{pers}} = 6.30$ and 4.8σ , and rules out any carrier above 0.66 of the discovery amplitude. The nearest covering block is 2.05 h away, so this is largely a within-night test: it excludes a persistent emitter of that strength and says nothing about an intermittent one. **The disposition rests on the recurrence test and on nothing else,** which is the chain applied to every crossing in the survey.

What the recurrence statistic reads when nothing is there

The exclusion in Table 8 is one formula. Writing S and σ_d for the discovery amplitude and its error, and $\bar{a}$ and

$\bar{\sigma} = (\sum_i \sigma_i^{-2})^{-1/2}$ for the inverse-variance combination of the repeat readings at the predicted channel,

$$\sigma_{\text{excl}} = \frac{S - \bar{a}}{\sqrt{\sigma_d^2 + \bar{\sigma}^2}} = \frac{T_{\text{pers}} - T_{\text{rep}}}{\sqrt{1 + (T_{\text{pers}}/T_\star)^2}}, \quad (\text{B1})$$

with $T_{\text{pers}} = S/\bar{\sigma}$ and $T_{\text{rep}} = \bar{a}/\bar{\sigma}$, so every exclusion follows from three columns of that table and nothing else. Because the discovery amplitude carries its own error, no exclusion can exceed the strength of the discovery it is built from — and that is also why it is not the strongest statement available. An amplitude selected as the largest of 10^2 to 10^6 cells at $T_\star \simeq 5$ is biased high, so a real carrier would be fainter than measured, and the quantity that survives the bias is the limit $(T_{\text{rep}} + 5)/T_{\text{pers}}$: the fraction of the discovery amplitude still ruled out at the trigger.

The statistic read at the predicted channel is centred on zero and scatters by one, measured and not merely designed. Over 511 single-cell readings of every repeat block in the census and the hold-out its mean is +0.014 and its scatter 1.046; at the neighbouring channels of the 2 crossings whose repeat spectra survive — random cells, same registration, same drift — +0.071 and 0.966 over 638 cells, with 3.19 the largest value anywhere within $\pm 15 \text{ km s}^{-1}$ of their predicted channels; and at the control positions of the discovery windows -0.036 and 1.034. The largest of the 8 retained control probes of each repeat block averages 1.449 against the 1.424 expected for a maximum of 8, which is the point: **a maximum is positive on average whatever is there.** The largest of the n_{rep} per-block readings averages +1.137 over the crossings carrying exclusions, where the null above predicts +1.148 for the same n_{rep} , while the combined T_{rep} of Eq. B1 averages -0.103. Widening the window costs trials and not calibration: the 28 exceedances in the 319 covering windows match the 27 the same accounting predicts, and HD 69830’s windows, at $\pm 50 \text{ km s}^{-1}$, are the only ones too narrow for the excursion their crossing’s drift implies.

THE LINE MASK

A crossing that falls on a known molecular transition has an astrophysical explanation to hand, and the mask exists to say which crossings those are. It is a label and not a verdict: every crossing goes on to the spatial and recurrence tests whether or not it carries the label, and the labelled crossings are tested below.

The list, and when it was made. The mask holds every transition of the 12 species these spectral setups cover — the three CO isotopologues, HCN, HCO⁺, CS, CN, SiO, H₂CO, SO, [C I] and hydrogen recombination — that falls inside one of the 46 searched frequency intervals with an upper-state energy below 150 K. That rule is the whole of it, and it admits 111 transitions (Table 7). No strength cut is applied across species, because the Einstein coefficients of the CO ladder are 10^{-7} – 10^{-6} s^{-1} and any threshold admitting the strong lines of HCN would discard the brightest lines in these fields; within a species the coefficient keeps a rotational transition with its main hyperfine components and drops the weak satellites. The rarer isotopologues, ³⁴SO and H¹³CN and the rest, are not among the named species and would add 525 transitions. The list is a single Splatalogue query of CDMS and JPL (Pickett et al. 1998; Müller et al. 2005) made

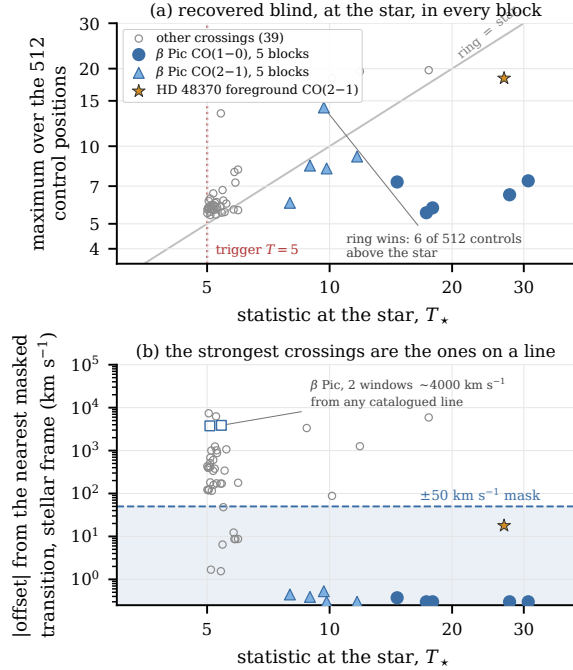


FIG. 10.— The β Pictoris positive control, in both directions. (a) The spatial screen. One point per execution block in which the blind search recovers circumstellar carbon monoxide, plotted as the stellar statistic against the maximum over that window’s 512 control positions: the 9 blocks in which the star outranks every control, and one further block in which the ring wins. *The control is that the search recovers a real narrow line at a stellar position, in two transitions and in every one of these blocks.* It is not that the star outranks the ring everywhere. In block X7116f1.X1785, Band 6, the control ring reaches 14.13 against 9.68 at the star, with 6 of the 512 control positions above it, so the screen rejects a window holding carbon monoxide that is real, centred on the star and independently mapped. The disc is resolved on the scale of the primary beam in that configuration, which is the geometry the screen is built to reject, so **the screen measures compactness and not authenticity** (§5.4). (b) Line confusion, which is what makes the screen’s job hard. Each crossing is plotted by its velocity offset from the nearest masked transition *in its own star’s rest frame*, with the adopted $\pm 50 \text{ km s}^{-1}$ half-width marked. The 10 β Pictoris carbon-monoxide crossings land $0.1\text{--}0.5 \text{ km s}^{-1}$ from CO, at the resolution of the systemic velocity itself, while the two CS($5\rightarrow 4$) crossings of the same star sit $3765\text{--}3870 \text{ km s}^{-1}$ from anything catalogued and are not attributed at any width. 39 further crossings are drawn in grey, with HD 48370 marked; 52 of the 56 are plotted, the remainder being crossings for which the release stores no control ring. *A crossing is identified by where it falls in this plane and not by how strong it is*, which is why the line mask and not the trigger is the productive test.

on 2026 October 6 and frozen with the data release, so it is reproducible as a file rather than as a description — and it therefore post-dates the 2026 September 11 fixing of the disposition criteria, as does every species in it. It is a uniform rule applied to all 56 crossings, not a record of what was recognised when each was first examined.

The frame is the star’s own, and the sign is not the usual one. A crossing’s frequency is carried from the topocentric value to the barycentre using its own block’s pointing and mid-time, and then to the stellar rest frame using the star’s systemic velocity. The barycentric term runs from -29.2 to $+29.2 \text{ km s}^{-1}$ across the survey, so a tolerance applied in the observed frame would spend 58 km s^{-1} of its budget on the Earth’s motion. The 3 HD 285968 epochs show the transform working in the only way that can be checked: 12 km s^{-1} apart as observed,

TABLE 7

THE LINE MASK, BY SPECIES: THE NUMBER OF TRANSITIONS N IT HOLDS, THE FREQUENCY RANGE THEY SPAN INSIDE THE SEARCHED INTERVALS AND THEIR UPPER-STATE ENERGIES. HYDROGEN RECOMBINATION LINES CARRY NO TABULATED E_u . THE TRANSITION-BY-TRANSITION LIST IS IN THE DATA RELEASE.

Species	N	ν_0 (GHz)	E_u (K)
H	5	92.03–231.90	
SO	15	100.03–682.68	16–143
H ₂ CO	13	101.33–491.97	10–106
C ¹⁸ O	3	109.78–329.33	5–32
CN	50	113.12–680.30	5–114
CO	3	115.27–345.80	6–33
CS	3	195.95–342.88	24–66
SiO	2	217.10–347.33	31–75
¹³ CO	8	220.40–330.59	16–32
HCN	7	354.50–354.51	43–43
HCO ⁺	1	356.73–356.73	43
[C I]	1	492.16–492.16	24

they agree to 0.03 km s^{-1} in the star’s frame. Throughout, $\Delta v_* = c(\nu_* - \nu_0)/\nu_0$, so a positive offset means the crossing lies *above* the transition in frequency — the opposite sign to the velocity convention of a spectral line.

The half-width, and what it costs. A one-channel test is too tight for circumstellar gas, whose emission is shifted by many channels: the tolerance follows from Keplerian motion where these molecules survive, $\lesssim 13 \text{ km s}^{-1}$ beyond 10 au, and is $\pm 50 \text{ km s}^{-1}$. At that width the mask removes 5.04 GHz of the 118.1 GHz unique-frequency union, which is 4.3 per cent of it; the loss is concentrated in the line-tuned fine-channel windows, and the cost band by band is in the data release. Between ± 20 and $\pm 100 \text{ km s}^{-1}$ the mask labels 19 to 22 of the 56 crossings, and the 1 unattributed crossing that outranks all 512 control positions in its own window is the same one at every width.

Sulphur monoxide decides three coincidences, and one rule is applied to all of them. CD–57 1054, CP–72 2713 and HD 161868 lie -1.7 , -12.3 and -48.0 km s^{-1} from sulphur monoxide transitions of upper-state energy 21–87 K, each inside a Band 7 window that also covers CO($3 \rightarrow 2$), and none has a nearer transition of any other masked species. There is reason for caution about each: sulphur monoxide has not been reported in a debris disc, where the second-generation gas is CO, C and O, and toward CP–72 2713 the CO($3 \rightarrow 2$) line is covered in the same window and absent to 10.5 mJy in a 0.42 km s^{-1} channel. That caution applies equally to each of them, and so does the date, since the species list post-dates every classification; neither separates one coincidence from the others. Applying the rule attributes all of them and gives 21 attributed crossings against 35 unattributed; withholding sulphur monoxide from all of them instead gives 18 and 38, and returns a second crossing to the 2 that outrank every control in their own windows. The recurrence test says the same thing about each of them either way, so the choice moves counts and no conclusion.

The labelled crossings are tested for recurrence too. Of the 21 attributed crossings, 18 have a later execution block covering the predicted cell. 13 are present again: toward β Pic and HD 285968 a later block carries its own threshold crossing at the same stellar-frame frequency, at $T_* = 5.9$ to 30.8 , which is the test working in the positive direction on emission that is certainly real. The other 5 are not present again, and in each the repeat coverage is

deep enough that a carrier at the discovery flux would have exceeded the trigger there and does not; among them are the sulphur monoxide coincidences, which the test limits to $4.1\text{--}5.5\sigma$ whichever side of the mask they are placed. 3 cannot be tested and are named rather than counted: HD 139084 B, HD 48370 and HIP 10679, each observed once at the frequency concerned.

Image-sideband leakage does not account for any crossing. The mask holds signal-sideband frequencies, so a bright line in the image sideband could in principle be mistaken for a carrier. 2 unattributed crossings invite the question, at 241.75 and 241.84 GHz toward β Pic, because each shares a block with a CO($2 \rightarrow 1$) crossing at 230.52 GHz in the opposite sideband. A mirror about the first local oscillator would place that line on the crossing only for an oscillator at 236.13–236.18 GHz, and the block’s own spectral windows require 235.02–235.68 GHz: at the setting the mirror needs, the lowest window of the block would sit at an intermediate frequency of 10.5 GHz, beyond the 10.0 GHz the receiver provides (Cortes et al. 2024). Their image frequencies fall at least 1353 km s^{-1} from every masked transition.

A wider list is not a better one. Every Splatalogue transition over the searched intervals under the same energy cut and with an Einstein coefficient above 10^{-5} s^{-1} — 5,114 transitions of 235 species — masks 75.1 per cent of the union and moves 27 of the 56 crossings across the line, one of them a crossing that outranks every control, at -36 km s^{-1} from a sulphur dioxide transition. At these frequencies most frequencies lie within $\pm 50\text{ km s}^{-1}$ of something in such a list, which is why the species are restricted to what these spectral setups cover, and why the outcome is a label. “Coincident with a known molecular line” is a statement about a catalogue rather than a property of a frequency.

THE FALSE-ALARM EXPECTATION AND THE CONTROL ENSEMBLE

The paper’s central claim is that the number of unattributed crossings is the number chance predicts. This appendix gives that expectation, what it is conditioned on, and what the control ensemble calibrating it can support.

What the expectation is conditioned on

The expectation is built from the control positions themselves, which are the only empirical null the data provide, and it imposes the condition an event must actually satisfy: to reach the trigger, and to fall outside the attribution windows. Each of the 402 Class A windows of the census contributes the fraction of its own 512 control positions that reaches the trigger, which is the probability that a position in that field does so — no cell count enters it and no form is assumed for the tail. Summed over the windows, 36.1 are expected to trigger by chance; the attribution windows occupy 4.3 per cent of the searched band; and the product is **32.8 expected unattributed Class A crossings against 29 observed**, the two sides taken over the same windows and under the same condition.

The one assumption that cannot be read off the searched data is that a star behaves like a control position in its own field, and the reserved blocks measure that in the quantity used here. Their stars reach the

trigger in 8 of 62 fine-channel windows where the control positions of those same windows predict 7.5: a ratio of 1.07, with 0.48–1.79 allowed by resampling the reserved blocks, and no two of the events in one block. In the 253 coarse-channel reserved windows no star reaches the trigger, against 0.04 predicted. No departure is detected and none is applied, and the limit is the one so few events allow: a departure smaller than about 35 per cent would not have been seen, and that is the accuracy of every chance expectation in this paper.

Every Class B crossing in the survey comes from the one execution block of §D.2. Setting that block aside, the coarse class adds almost nothing to either side — the expectation over both classes together is 32.9 against the same 29 observed — so the Class A comparison is also the survey’s. Only 455 of the 1651 windows hold any control position above $T_{\text{trig}} = 5$ at all, and 398 of those are Class A: a coarse single-pointing window has too few cells for its own field to reach the trigger by chance.

One set of windows was extracted twice, and the second extraction is the one adopted. 157 windows were first extracted over a field that excluded integrations lying inside the primary beam; re-extracting them recovers a median 2.1 times the on-source time and shifts the search statistic by a median 0.07, or 0.36 in absolute value. The rule for which integrations a window keeps was fixed before any statistic in this paper was computed, and the re-extracted value is adopted wherever it exists: it carries 8 rows of Table 8 and takes 4 crossings below the trigger. A re-extracted window’s control ensemble is re-extracted with it, so neither side of the comparison above mixes the two. The deposited catalogue holds the first extraction’s statistic and the shorter species list in its own columns, so its disposition column and the adopted ledger disagree on those 4 crossings and on 3 more that the species list now attributes. The ledger is the authority for every disposition quoted here.

Execution-block quality

The comparison with the control positions assumes each block’s ensemble behaves like noise, and one block makes that assumption plainly false: every one of its 512 probes exceeds the trigger, the smallest at 6.6. A criterion for the condition was therefore fixed and applied to every block, rather than recognised on the block that prompted it.

The statistic is the one this paper already uses as a diagnostic: the median of a window’s control ensemble, which is what separates a field with a source in it from a field that is no null at all. That median is a maximum over the window’s own channel $\times$ drift grid, so its level rises with the cell count; it is therefore scaled by the median of the same quantity over the windows of its own search class, and one threshold then means the same thing in both. A block’s statistic is the median of that ratio over its windows, and **a block fails when it exceeds twice the scale of its class** — the factor already carried by the window-level data-quality flag, fixed before that flag was applied to anything.

Applied to all 404 blocks of the primary census, where the median block sits at 1.00, the criterion fails 1: A002_Xc7fa6f_X28a8 toward BD+05 1668, at 4.1 where the next worst block is 1.12. Nothing else is close, and the unscaled form of the rule fails the same 1 block. Its

4 windows are all coarse-channel and hold the survey’s entire Class B crossing population, so the block carries 4 of the 56 raw threshold crossings and no Class A statistic in this paper has ever contained it. **The population the paper characterises is therefore the 52 crossings in quality-passing data:** 21 attributed, 31 unattributed, 1 of those outranking all 512 control positions of their own window against 0 in the failed block. The other 4 appear in Table 8, with their block named, and enter no count anywhere else.

The control ensemble: its calibration, and its resolution

The ring geometry is set in §4.1; its calibration and its resolution are the subject here. Under that geometry the stellar ranks are consistent with the uniform distribution exchangeability implies, in both array strata and over the 1645 windows that retain a control vector, the ACA windows measured on the geometry of each observation’s own antenna table and baselines rather than on a 12 m beam. The test is not a powerful one: a shift of less than about 0.042 in the rank distribution could not have been seen in samples of this size, so the strata are consistent with uniform rather than demonstrably uniform.

The rank is not a significance test, and the reason is structural. Ranking the star against 512 control positions drawn from the same field is a good way to decide which windows deserve examination, but $1/(512 + 1) = 1.95 \times 10^{-3}$ is the *resolution* of the ensemble — the smallest rank it can express — and not the probability that noise produced the window. A survey of 1651 windows needs $0.05/1651 = 3.0 \times 10^{-5}$, so that floor is 64 times too coarse, and correlation widens it: neighbouring controls are correlated below the synthesised-beam scale, so a window’s annulus carries only $N_{\text{eff}} \simeq 30\text{--}43$ independent spatial trials and the ensemble resolves no better than 0.023–0.032. Drawing more controls does not rescue it, the primary beam bounding how many independent positions a single pointing contains. **That floor is the design’s central limitation: no single window in this survey can be significant on the rank alone, which is why confirmation here requires independent recurrence.**

The trials budget

The searched cells counted in §4.2 are not independent trials: channel smoothing and a drift grid stepped twice per channel correlate neighbouring ones, which alone would leave 0.19 of them independent. The control ensembles say otherwise. Their maxima sit at 0.995 of the prediction for the maximum of the full searched count treated as independent and Gaussian, a budget that predicts 476 windows carrying a control above T_{trig} against the 455 observed; and fitting each window’s effective count against its own ensemble returns 0.81 of its searched cells. Nothing quoted here turns on which figure is right, because the expectation above is summed from the measured exceedance rates and not from a cell count; by the cell-count route it is 36.5, 3 per cent away.

The on-star crossings do exceed that budget, and the attribution accounts for it. A Gaussian tail over every cell the survey searched predicts 37.8 stellar exceedances against the 56 found, but 21 of those are identified with a catalogued transition and 4 come from the block of §D.2;

what is left is the 29 unattributed crossings in quality-passing data against 32.9 predicted. No excess of chance crossings survives the attribution, which is the comparison the paper rests on.

One stratum departs from uniformity and it is the one that matters. The quantity is the number of windows whose stellar peak both reaches the trigger and exceeds all 512 of that window’s own control positions, and over $n = 402$ fine-channel windows it is not uniform: 12 such windows against 0.78 expected, while the coarse stratum, three times larger and predicting 2.43, contains 0 — including none of the 4 crossings of the flagged block, whose statistics stand below their own re-extracted control maxima. A coarse channel dilutes a narrow circumstellar line, so the class that can resolve one is the class that finds it — and by the same reasoning a genuine unresolved carrier would appear in Class A first. The departure survives removal of every fine window carrying a stellar crossing, so fine-window rank probabilities are optimistic for the class as a whole and not only where a crossing sits.

What is left is bounded, and the bounds are null. Treating each control as a pseudo-star and ranking it against the other 512-1 gives 220672 ranks that the real stellar ranks agree with at $p = 0.38$; any systematic radial suppression of the controls is below 10 per cent in the mean; and scrambling each window so that nothing coherent survives along a drift track carries 4 windows past the rank floor and changes no disposition.

RADIO-FREQUENCY INTERFERENCE SCREENING

Interference is the first alternative explanation for an unattributed crossing, so the screening this survey has, and the coverage it does not have, are both set out here. Millimetre interferometry is an unusually poor environment for terrestrial interference, though not an empty one: ITU Radio Regulations Article 5 (International Telecommunication Union 2024) carries active allocations throughout 90–275 GHz and WRC-19 identified parts of 275–450 GHz for land-mobile and fixed use. The specific hazard is the 94.0–94.1 GHz Earth exploration-satellite allocation used by spaceborne cloud radar (Stephens et al. 2002; Wehr et al. 2023), a documented contaminant of ALMA Band 3; 1 of the 1651 searched windows overlaps it, carrying 0 threshold crossings and 0 rank-passing events, so no result here rests on it. What favours this band is that terrestrial transmitters are sparse compared with centimetre wavelengths, that the array sits at 5000 m on a site chosen partly for radio quietness, and, decisively, that an interferometer is not a total-power detector: a signal not in the far field at the phase centre does not fringe-track but acquires a rapidly varying geometric phase across the array, and is suppressed in the correlation rather than imaged.

The coverage of the screening is uneven, and that is stated rather than smoothed over. The allocation test above and the sky- and intermediate-frequency tests below are computed for every searched window from the released catalogue, as is the 512-position rank screen, which asks whether an excess is at the star or everywhere in the field. Recurrence reaches only the crossings with a covering repeat block, and the visibility fit is a consistency check rather than a filter. The correla-

TABLE 8

THE CROSSING LEDGER, FROM WHICH EVERY DISPOSITION IN THIS PAPER CAN BE REPRODUCED. ALL 56 THRESHOLD CROSSINGS, THE 4 IN THE FLAGGED BLOCK OF §5.3.2 AMONG THEM AND LABELLED AS SUCH: THE STAR, ITS EXECUTION BLOCK, THE ALMA BAND, THE CROSSING FREQUENCY AS THE CORRELATOR DELIVERED IT AND THE SAME FREQUENCY IN THE STAR'S REST FRAME, THE SEARCH STATISTIC T_* , THE LARGEST STATISTIC T_{ctrl} AMONG THE WINDOW'S 512 CONTROL POSITIONS, THE NEAREST MASKED TRANSITION AND THE STELLAR-FRAME OFFSET Δv_* FROM IT, THEN THE RECURRENCE TEST: n_{rep} COVERING REPEAT BLOCKS SPANNING Δt DAYS, THE STATISTIC T_{pers} A CARRIER PERSISTING AT THE DISCOVERY FLUX WOULD HAVE PRODUCED AT THE PREDICTED STELLAR-FRAME CELL, THE STATISTIC T_{rep} MEASURED THERE AT THE CARRIED DRIFT, COMBINING THE REPEAT EPOCHS BY INVERSE VARIANCE, AND THE EXCLUSION $\sigma_{\text{excl}} = (T_{\text{pers}} - T_{\text{rep}})/\sqrt{1 + (T_{\text{pers}}/T_*)^2}$ (Eq. B1), WHICH IS THEREFORE REPRODUCIBLE FROM THREE COLUMNS OF THIS TABLE. THE FRACTION OF THE DISCOVERY AMPLITUDE STILL RULED OUT AT THE TRIGGER, $(T_{\text{rep}} + 5)/T_{\text{pers}}$, FOLLOWS FROM TWO OF THEM. T_* AND T_{ctrl} ALWAYS COME FROM THE SAME EXTRACTION OF THE SAME WINDOW, SO THE COMPARISON BETWEEN THEM IS ONE A READER CAN MAKE; T_{ctrl} GATES NOTHING AND SAYS ONLY WHICH CROSSING WAS EXAMINED FIRST, AND THE DRIFT-MAXIMISED READING OF THE CELL IS IN TABLE 4 FOR THE TWO LEADING EVENTS AND IN THE DATA RELEASE FOR THE REST. ON THE FLAGGED ROWS T_{pers} AND THE EXCLUSION, WHICH WOULD INHERIT A DISCOVERY AMPLITUDE FROM THE FAILING BLOCK, ARE WITHHELD. A CROSSING IS ATTRIBUTED WHERE $|\Delta v_*| \leq 50 \text{ km s}^{-1}$ AND THE TRANSITION NAMED IS THE NEAREST IN THAT FRAME, SO FAR FROM EVERY TRANSITION IT IS THE MAGNITUDE AND NOT THE NAME THAT CARRIES INFORMATION. BLOCK IDENTIFIERS OMIT THE CONSTANT A002_ PREFIX.

Star	block	$B \nu_{\text{topo}}$	$(\text{GHz}) \nu_*$	T_*	T_{ctrl}	line	$\Delta \nu_*$	n_{rep}	Δt	(d)	T_{pers}	T_{rep}	σ_{excl}	disposition
BD+05 1668	Xc7fa6f_X28a8	6	240.1016	240.1054	73.398	74.69 CS(5-4)	-5911.9	290.9-113	n/a+0.15	n/a	unattributed; flagged			
BD+05 1668	Xc7fa6f_X28a8	6	224.7422	224.7458	48.303	83.74 H ₂ CO(3 _{1,2} -2 _{1,1})	-1264.5	290.9-113	n/a+0.37	n/a	unattributed; flagged			
BD+05 1668	Xc7fa6f_X28a8	6	226.7422	226.7458	41.355	75.48 CN(2 _{3/2} -1 _{1/2})	+88.0	290.9-113	n/a+0.66	n/a	unattributed; flagged			
BD+05 1668	Xc7fa6f_X28a8	6	242.1953	242.1992	39.586	83.29 CS(5-4)	-3349.2	290.9-113	n/a+0.77	n/a	unattributed; flagged			
β Pic	Xf5d76d_Xeb8	3	115.2603	115.2711	30.765	7.34 CO(1-0)	-0.2	—	—	—	—	—	—	attributed
β Pic	Xf5d76d_Xe19	3	115.2603	115.2711	27.684	6.48 CO(1-0)	-0.2	—	—	—	—	—	—	attributed
HD 48370	Xc26103_X155a	6	230.5117	230.5243	26.834	18.36 CO(2-1)	-17.8	—	n/a	—	n/a	attributed; flagged		
β Pic	Xf4df6f_Xc7	3	115.2620	115.2711	17.907	5.76 CO(1-0)	-0.2	—	—	—	—	—	—	attributed
β Pic	Xf5d76d_Xcc5	3	115.2604	115.2711	17.288	5.52 CO(1-0)	-0.2	—	—	—	—	—	—	attributed
β Pic	Xf5d76d_X32f1	3	115.2606	115.2713	14.654	7.27 CO(1-0)	+0.4	—	—	—	—	—	—	attributed
β Pic	Xd9668b_X3a90	6	230.5161	230.5378	11.681	9.12 CO(2-1)	-0.3	—	—	—	—	—	—	attributed
β Pic	X6f1341_X1484	6	230.5284	230.5379	9.832	8.20 CO(2-1)	-0.1	—	—	—	—	—	—	attributed
β Pic	X7116f1_X1785	6	230.5272	230.5384	9.677	14.13 CO(2-1)	+0.5	—	—	—	—	—	—	attributed
β Pic	Xd9668b_Xa9df	6	230.5160	230.5377	8.948	8.41 CO(2-1)	-0.4	—	—	—	—	—	—	attributed
β Pic	Xa7a216_X23e4	6	230.5273	230.5377	7.988	6.03 CO(2-1)	-0.4	—	—	—	—	—	—	attributed
61 Vir	Xc079b5_X82f	7	346.3138	346.3232	5.958	5.65 SO(9 ₈ -8 ₇)	-177.6	3	0.1-3.0	9.7+0.39	4.9	unattributed; no recurrence		
HD 285968	Xd98580_X3b6f	6	230.5021	230.5447	5.956	8.12 CO(2-1)	+8.7	—	—	—	—	—	—	attributed
HD 285968	Xdb7ab7_X5b4e	6	230.5116	230.5447	5.868	7.22 CO(2-1)	+8.7	—	—	—	—	—	—	attributed
HD 285968	Xdb4b9a_X108f	6	230.5100	230.5447	5.835	7.94 CO(2-1)	+8.7	—	—	—	—	—	—	attributed
CP-72 2713	Xf0235_X4a6d	7	344.2697	344.2965	5.809	5.68 SO(8 ₈ -7 ₇)	-12.3	5	0.1-2.1	6.3 -0.82	4.8	attributed		
HD 139084	B Xcd8029_Xb6b0	7	345.8241	345.8241	5.763	17.70 CO(3-2)	+24.4	—	—	—	—	—	—	attributed
HD 14055	Xff3d8_Xd4e	7	331.7699	331.7664	5.575	5.99 ¹³ CO(3-2)	+1068.7	300.9-650	170.4 -0.02	5.6	unattributed; no recurrence			
HD 23484	X10b22d7_X2445	6	230.7904	230.8006	5.520	5.55 CO(2-1)	+341.5	4	0.1-2.1	10.4+1.25	4.3	unattributed; no recurrence		
HD 161868	Xff45a6_X10ca8	7	345.6338	345.6491	5.481	6.16 SO(2 ₃ -2 ₁)	-48.0	13	0.0-6.1	92.2+0.24	5.5	attributed		
HIP 10679	Xa95c04_X14cd	6	220.4185	220.4035	5.456	6.76 ¹³ CO(2-1)	+6.5	—	—	—	—	—	—	attributed
HD 14055	X11adad7_X25c5e7	7	345.4564	345.4303	5.448	6.82 SO(2 ₃ -2 ₁)	-237.8	300.1-653	14.0+0.03	5.1	unattributed; no recurrence			
HD 207129	Xc19d6f_X5706	6	230.4226	230.4070	5.433	5.52 CO(2-1)	-170.3	40.8-425	69.7 -0.93	5.5	unattributed; no recurrence			
β Pic	Xd9668b_X3a90	6	241.7512	241.7739	5.420	5.81 CS(5-4)	-3869.7	1	1.0	5.7+0.72	3.4	unattributed; no recurrence		

TABLE 9

The crossing ledger, continued. THE REMAINING 28 OF THE 56 THRESHOLD CROSSINGS, IN THE SAME ORDER AND WITH THE SAME COLUMNS AS TABLE 8, WHOSE CAPTION DEFINES THEM.

Star	block	B	ν_{topo}	(GHz)	ν_*	T_*	T_{ctrl}	line	Δv_*	n_{rep}	Δt (d)	T_{pers}	T_{rep}	σ_{excl}	disposition
HR 1010	Xc6fa08_X1053	6	230.5133	230.5287	5.410	5.65	CO(2-1)	-12.0	—	—	—	—	—	—	attributed
ALMA J153702653-33192492	Xff0235_X8392	6	230.5222	230.5392	5.405	13.41	CO(2-1)	+1.6	—	—	—	—	—	—	attributed
HD 110411	Xe45e29_Xb7d4	6	219.4538	219.4357	5.393	5.85	C ¹⁸ O(2-1)	-170.2	1	10	6.7+0.62	3.8	unattributed;	no recurrence	
HD 31392	X129754b_X6946	6	231.1998	231.2231	5.289	5.81	H(30a)	-876.3	8	0.1-41	15.7 -0.26	5.1	unattributed;	no recurrence	
HD 14055	Xff99e1_X24be0	7	331.7078	331.7035	5.277	5.69	¹³ CO(3-2)	+1011.6	30	0.9-652	252.2+0.15	5.3	unattributed;	no recurrence	
AU Mic	Xa42f75_X319	6	230.6864	230.6705	5.274	6.07	CO(2-1)	+172.3	2310-455	6.4 -0.16	4.2	unattributed;	no recurrence		
HD 69830	X12209f7_X16edb7	7	322.4325	322.4444	5.259	5.42	C ¹⁸ O(3-2)	-6268.5	5	0.1-29	8.9 -0.48	4.8	unattributed;	no recurrence	
ALMA J153702653-33192492	Xfe62c1_X398	6	216.1844	216.2036	5.228	6.02	SiO(5-4)	-1244.6	7	18-126	11.0+1.03	4.3	unattributed;	no recurrence	
AU Mic	X899169_X1838	6	230.8252	230.8294	5.221	5.85	CO(2-1)	+379.0	2145-310	10.0+0.19	4.5	unattributed;	no recurrence		
HD 14055	Xff99e1_X2c3bb	7	345.5256	345.5216	5.186	5.66	SO(2 ₃ -2 ₁)	-158.6	30	0.1-651	172.8+2.00	5.1	unattributed;	no recurrence	
HD 14055	Xfffdel_X3835	7	345.0099	345.0070	5.177	6.56	SO(2 ₃ -2 ₁)	-604.8	30	0.1-649	187.2 -1.78	5.2	unattributed;	no recurrence	
HD 14055	Xff99e1_X2b2ee	7	344.7007	344.6965	5.175	5.92	SO(8 ₈ -7 ₇)	+336.0	30	0.1-651	210.7 -0.00	5.2	unattributed;	no recurrence	
LHS 1140	Xfd9d7c7_X6fd2	6	230.6355	230.6268	5.136	5.68	CO(2-1)	+115.5	4	5.0-27	7.3+0.33	4.0	unattributed;	no recurrence	
HD 10647	Xff294f_Xa7f0	7	331.6570	331.6943	5.120	5.91	¹³ CO(3-2)	+1003.3	10	0.1-592	125.8+0.39	5.1	unattributed;	no recurrence	
CD-57 1054	Xc5f525f_X5aa4	7	346.5099	346.5265	5.108	5.71	SO(9 ₈ -8 ₇)	-1.7	1	39	6.0 -0.27	4.1	attributed		
ALMA J153702653-33192492	Xf8f6a9_X10359	6	217.7292	217.7269	5.091	5.41	H ₂ CO(3 _{0,3} -2 _{0,2})	-680.5	7	3.8-242	11.6+0.07	4.6	unattributed;	no recurrence	
GJ 849	Xd9668b_X841a	6	230.8769	230.8581	5.090	5.83	CO(2-1)	+416.3	5	0.1-51	9.3 -1.69	5.3	unattributed;	no recurrence	
ALMA J153702653-33192492	Xfe62c1_X398	6	218.8729	218.8923	5.090	5.56	H ₂ CO(3 _{2,1} -2 _{2,0})	+181.2	7	18-126	10.5 -1.04	5.0	unattributed;	no recurrence	
HD 14055	X1001fde_X29ae	7	331.1559	331.1546	5.082	6.26	¹³ CO(3-2)	+513.8	—	—	—	—	unattributed;	no spectrum	
β Pic	Xd9668b_Xa9df	6	241.8367	241.8594	5.079	5.99	CS(5-4)	-3765.1	1	1.0	4.8 -0.63	3.9	unattributed;	no recurrence	
ALMA J153702653-33192492	Xf8f6a9_X10359	6	218.6711	218.6688	5.048	5.77	H ₂ CO(3 _{2,1} -2 _{2,0})	-125.1	7	3.8-242	11.6+0.00	4.6	unattributed;	no recurrence	
HD 69830	X12277ed_X178c6	7	321.2005	321.2149	5.046	6.13	C ¹⁸ O(3-2)	-7387.7	5	6.0-36	10.2 -0.46	4.7	unattributed;	no recurrence	
HR 1010	Xc66ce1_Xaa9	6	230.8348	230.8491	5.044	5.72	CO(2-1)	+404.6	7	1.1-14	10.3 -1.53	5.2	unattributed;	no recurrence	
61 Vir	Xc067f7_X822	7	345.5599	345.5678	5.038	5.79	SO(2 ₃ -2 ₁)	-118.5	3	2.0-3.0	9.3+0.45	4.2	unattributed;	no recurrence	
HD 170773	Xff0235_X8c42	7	331.0101	331.0234	5.029	5.75	¹³ CO(3-2)	+394.8	4	0.9-113	74.8+1.56	4.9	unattributed;	no recurrence	
ALMA J153702653-33192492	Xd9d7c7_Xbb64	7	347.2179	347.1890	5.009	5.68	SiO(8-7)	-122.2	1	0.1	5.0 -2.06	5.0	unattributed;	no recurrence	
HD 14055	X1190de8_Xd526	7	344.4676	344.4515	5.006	5.96	SO(8 ₈ -7 ₇)	+122.7	30	20-611	11.6 -0.60	4.8	unattributed;	no recurrence	
η Crv	X122b6ff_X1041e	7	357.2815	357.2475	5.006	5.50	HCO ⁺ (4-3)	+431.3	1	87	11.0 -0.62	4.8	unattributed;	no recurrence	

tor flagging is the observatory's own, replayed from each block's delivered calibration script, but it is not recorded per window in the release. *No per-channel interference flag of our own exists, and none can be recovered.* Three measurements follow, and each is a test the population could have failed.

First, clustering in sky frequency. If interference dominated the crossing population, the crossings would pile up at particular sky frequencies. They do cluster — but

on the survey's own molecular lines. The test needs a measured peak frequency in the frame in which an interfering carrier would stand still, so it runs topocentrically over the 51 of the 56 crossings for which the release records one, against the transitions the search was tuned to. For the other 5, all β Pic carbon-monoxide crossings, the release gives the velocity offset from the line instead, and a frequency reconstructed from a line offset cannot be used to ask whether frequencies crowd onto lines. Of

the 51, 11 fall within $\pm 50 \text{ km s}^{-1}$ of a searched transition against 4.7 expected if each window's peak fell anywhere on that window's own channel grid, $p = 0.0007$. The densest band, 230.422–230.723 GHz, holds 12 of them, 9 within that velocity window of CO(2→1); and once the line-attributed crossings are set aside, the 40 that remain show no clustering at all ($p = 0.111$). *The concentration is the attribution, not a residue left over after it.* Table 8 counts the same band as 14 crossings with 11 attributed, because it is a different test over a different set — all 56, each attributed in its own star's rest frame — and the 2 it adds are exactly the ones whose peak frequency this test has no measurement of.

Second, fixed intermediate frequency. An artefact of the signal chain would sit at a fixed position within the spectral window rather than at a fixed sky frequency. This test is run over every window for which the catalogue releases a peak frequency — 1397 of the 1651, of which only 51 carry a threshold crossing — because such an artefact would occupy the same intermediate frequency in all of them and not only where the trigger fires. It does not: the peaks' fractional positions within their own windows have median 0.50 and are uniform.

Third, a cross-target occupancy screen over every searched window and not only the crossings. Interference belongs to the observatory and not to the star, so a terrestrial carrier should recur at the same sky frequency toward *unrelated* stars. It does not even for the leading event: the one unattributed rank-passing event has a released crossing frequency; 40 further windows, toward 24 other stars in 32 other blocks, cover that frequency, and none of them carries a crossing in the same channel. Over all 3027 searched windows toward 133 stars, with each window's peak randomised on its own channel grid, the number of cross-star coincidences within 5 MHz is 569 against 554 ± 13 expected ($p = 0.15$); among the 109 windows whose peak exceeds the trigger it is 4 against 5.2 ± 1.9 , i.e. *below* the null, and with the molecular-line mask applied 1 against 2.1 ± 1.3 . There is no excess to explain. Two features of the screen matter. It is keyed on the star and not on the product directory, because a star's own repeat observations are not different targets; and 35 of those 109 windows sit on a masked transition (CO(2→1) $\times 17$, CO(1→0) $\times 10$, CO(3→2) $\times 4$, $^{13}\text{CO}(2\rightarrow 1) \times 3$, SiO(5→4) $\times 1$), so the mask has to be applied first — a carbon monoxide line falls at nearly the same sky frequency toward every nearby star, which is exactly the coincidence the screen is looking for.

One execution block fails these tests, and it fails them as a block rather than as a carrier. Its 4 crossings toward

BD+05 1668 are the highest statistics in the survey outside β Pictoris and HD 48370, and the field statistics that identify the failure are in §5.3.2. What belongs here is whether interference accounts for it, and three readings say not. The visibility fit returns a large imaginary part, placing the emission away from the phase centre. The block's 24 sibling executions of the same field, with the same array and the same spectral setup, produce no crossing at all. And it is not a carrier at a fixed sky frequency: each of the four frequencies is covered by 127–137 further windows toward 17–22 other stars, and 0 of them carries a crossing in the same channel.

One reading does point at a mechanism, and it can be tested across the survey. 2 of the 4 sit at the identical channel of two different spectral windows, which is the signature of a fixed intermediate frequency — and a feature with no geometric fringe phase, a spur common to every antenna being the obvious case, does not fringe-track, so it gathers at the phase centre rather than at the star. Here the two are a twentieth of an arcsecond apart, so the block cannot test its own hypothesis; the survey can, because the star lies 5.0–8.1 arcsec from the phase centre in 75 windows whose control annulus brackets it, putting a median of 15 probes within two synthesised beams of the phase centre and the star well outside. Against the probes at the same distance from the star but elsewhere in azimuth — matching the radius matters, or a centrally concentrated disc would imitate the effect — those probes are higher by $+0.09 \pm 0.08$ in the statistic, consistent with zero and bounded by 0.26, smaller by a factor of about 33 than the 8.4 by which this block's ring is raised. *The phase centre is not where the excess lives. The evidence supports a data-quality failure of one execution block, and that is how these 4 crossings are dispositioned* (§5.3.2): the criterion of Appendix D.2 flags 5 of the 1651 windows, toward 2 stars (BD+05 1668 and HD 48370), the second being a resolved disc rather than a defect.

None of this is a substitute for the ON/OFF cadence a dedicated search uses, and we do not present it as one. We claim only that *no identified interference mechanism accounts for the unattributed events*: allocation tables cannot exclude harmonics, intermodulation products, unconventional emitters or near-field sources, and archival data without a simultaneous off-source strategy cannot do better. The three tests establish that the crossing population as a whole carries none of the signatures interference leaves, which is a statement about the population and not a clearance of any individual crossing.

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